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Sparse feature selection using hypergraph Laplacian-based semi-supervised discriminant analysis

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ABSTRACT

Feature selection, as a dimension reduction technique in data mining and pattern recognition, aims to select the most discriminative features and improve the learning performance. With an abundance of unlabeled data readily available across various applications, semi-supervised feature selection has emerged as a promising approach. While most semi-supervised feature selection methods rely on simple graphs to preserve the geometrical structure of data, this approach often fails in capturing the high-order relationships present in many real-world applications. In contrast, hypergraphs offer the ability to encode more complex structures of data beyond what a simple graph can achieve. In this paper, we propose a feature selection method formulated in the trace ratio form, integrating hypergraph Laplacian-based semi-supervised discriminant analysis (SDA) and the mixed convex and non-convex $\ell_{2,p}$-norm ($0 < p \leq 1$) regularization. The proposed trace ratio-based method, called HSDAFS, leverages the discriminative information from labeled data to maximize class separability while also utilizing the hypergraph Laplacian to capture the geometrical structure and high-order relationships within both labeled and unlabeled data. The $\ell_{2,p}$-norm regularization in the proposed HSDAFS provides improved sparsity over the $\ell_{2,1}$-norm. It ensures that the projection matrix is row-sparse, enabling the effective joint selection of discriminative features across all data. To solve the trace ratio-based HSDAFS method, we convert it into a trace difference method and propose an iterative algorithm. Experiments on several datasets demonstrate that HSDAFS is more effective in selecting the most discriminative features compared to other methods.

1. Introduction

In a variety of applications, high-dimensional data are available that include redundant and noisy features. These features reduce learning performance and increase computational cost [1,2]. Feature extraction (FE) and feature selection (FS) are used as important techniques to reduce data dimensions and increase learning performance [1]. FE involves generating new features from original data by transforming them into a lower-dimensional space [3]. This process reduces the risk of overfitting, accelerates model training, and improves computational efficiency [4]. FS is a technique that addresses the problem of eliminating redundant, irrelevant, and noisy features, and selecting the important and discriminative features, leading to better and more accurate results [5,6]. In many real-world applications where irrelevant and noisy features are prevalent, FS can be a more effective technique than FE for reducing data dimensions and enhancing learning performance. FS can be especially valuable in applications where feature interpretability is essential [1,7]. Moreover, FS allows for the collection or computation of only the selected features, unlike FE, which requires all features to achieve a low-dimensional representation of data. Consequently, significant research efforts have been dedicated to addressing the FS problem in recent years [1].

FS methods are divided into filter, wrapper, and embedded methods [8]. Filters serve as a preprocessing step where features are evaluated independently of the classifier using statistical performance criteria [9]. Wrappers involve applying classification algorithms to assess feature performance [8,9]. In embedded methods, FS is integrated into the training process [8].

FS methods are further classified into supervised, unsupervised, and semi-supervised methods [10]. Unsupervised FS methods disregard label information and focus on identifying features that maintain the data distribution [11]. Supervised FS methods use labeled data to

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identify the most discriminative features for distinguishing between classes [12]. Due to the significant impact of discriminant analysis on supervised FS, linear discriminant analysis (LDA), a widely used supervised technique in FE, serves as the basis for some FS methods. LDA aims to find a projection matrix W that maximizes the between-class scatter matrix and simultaneously minimizes the within-class scatter matrix [13,14]. This dual objective can be expressed in the trace ratio form as $\max_W \mathrm{Tr}(W^T M^b W)/\mathrm{Tr}(W^T M^w W)$, where M^b and M^w represent the between-class and within-class scatter matrices, respectively. However, obtaining a closed-form solution for the trace ratio problem is difficult, and it is typically transformed into the ratio trace form $\max_W \mathrm{Tr}((W^T M^w W)^{-1}(W^T M^b W))$, which can be solved via generalized eigenvalue decomposition [14,15]. Some efficient algorithms have been developed to directly solve the trace ratio formulation by converting it to an equivalent trace difference problem [16]. Trace ratio-based LDA often outperforms ratio trace-based LDA in many classification tasks [15]. Therefore, our focus of this paper for FS is on the trace ratio form.

Many LDA-based FS methods evaluate each feature separately, disregard the relevance between different features, and fail to select the most informative feature combinations. To tackle this problem, sparse FS methods based on LDA have been proposed. These methods consider the feature relevance and jointly select features from all data points by incorporating sparse models as regularization terms to ensure row-sparsity in the LDA projection matrix [17]. This row-sparse projection matrix can then be utilized to rank the features and select the top-ranked ones. The $\ell_{2,1}$-norm and the mixed $\ell_{2,p}$-norm $(0 < p \leq 1)$ [18] are sparse models widely applied for sparse FS. The $\ell_{2,p}$-norm makes the problem convex when $p = 1$ and non-convex when $0 < p < 1$, often outperforming the $\ell_{2,1}$-norm.

Masaeli et al. [19] presented a discriminant FS that uses $\ell_{\infty,1}$-norm on the projection matrix of LDA. In this method, the possibility of arbitrary scalability of the projection matrix is ignored, and a trivial solution of all zeros is achieved. Therefore, it cannot select features at the trivial solution. Tao et al. [20] employed $\ell_{2,1}$-norm regularization on the formulation of LDA for making the matrix sparse in rows and suitable for FS. Then, they extended their formulation to the $\ell_{2,p}$-norm $(0 < p < 1)$ regularized case. The mentioned methods rely on labeled data and may fail to identify discriminative features when only a limited amount of labeled data is available [21]. However, collecting plentiful labeled data is often costly and requires the labor of human experts [22]. To mitigate these challenges, semi-supervised FS methods have been developed, which leverage both labeled and unlabeled data to better identify the most discriminative features while preserving the geometrical structure of data. Most semi-supervised FS methods utilize simple graphs to retain the geometrical structure of data. However, simple graphs only consider pairwise relationships between two data samples and ignore high-order relationships present in the original data. In many real-world scenarios, the relationships among data points exhibit more complexity than simple pairwise relationships. Hence, this complexity can result in a decrease in the efficiency of FS methods that utilize simple graphs. In contrast, hypergraphs extend the concept of simple graphs by allowing hyperedges to connect any number of vertices [23,24]. This extension enables hypergraphs to model the complex correlations within real-world data, facilitating the capture of high-order relationships among data points [24].

Given the challenges mentioned above, we propose an FS method using hypergraph Laplacian-based semi-supervised discriminant analysis (SDA) and $\ell_{2,p}$-norm $(0 < p \leq 1)$ regularization. SDA [25], an extension of LDA, is specifically designed for semi-supervised scenarios, with its objective function formulated in the ratio trace form. It constructs a projection matrix for FE by utilizing the graph Laplacian of a simple graph. Our proposed method, called Hypergraph Laplacian-based Semi-supervised Discriminative Analysis for Feature Selection (HSDAFS), reformulates the objective function of SDA into a trace ratio form for an FS framework. It leverages the $\ell_{2,p}$-norm regularization, the inter-class distance, and the geometrical structure of all data based on hypergraph Laplacian. The trace ratio-based HSDAFS method is designed to identify the most discriminative features while capturing the high-order relationships within the data. In the HSDAFS, the mixed $\ell_{2,p}$-norm regularization ensures that the projection matrix is well-suited for FS. Since the HSDAFS method comprises the mixed $\ell_{2,p}$-norm, it is convex for $p = 1$ and non-convex for $0 < p < 1$. Moreover, the HSDAFS method is in the trace ratio form. Accordingly, the optimization of the HSDAFS framework is not easy. In response to this challenge, a unified algorithm is proposed to solve our trace ratio-based method for the convex and non-convex cases. Experimental results on several datasets represent the effectiveness of the proposed HSDAFS in identifying the most discriminative features.

The contributions of this paper comprise:

- We propose a hypergraph Laplacian-based semi-supervised sparse FS method using the trace ratio formulation. This method integrates SDA and mixed $\ell_{2,p}$-norm regularization in the trace ratio form to construct a sparse projection matrix for FS. The $\ell_{2,p}$-norm regularization in the proposed trace ratio formulation makes it suitable for FS. By utilizing the discriminative capabilities of SDA and capturing the high-order relationships of data, our method demonstrates superiority over traditional graph-based semi-supervised FS methods.
- Our method leverages labeled data to maximize class separability and utilizes all available data to preserve the geometrical structure of data through hypergraph Laplacian. It effectively captures high-order relationships within the data.
- The proposed method, formulated in the trace ratio form, incorporates the mixed $\ell_{2,p}$-norm which is convex when $p = 1$ and non-convex when $0 < p < 1$. Consequently, the trace ratio-based method is transformed into an equivalent trace difference method, and a novel iterative algorithm is proposed to unifiedly solve the method for all $p \in (0, 1]$.
- The performance of our method is evaluated utilizing several datasets and compared with other FS methods. The results demonstrate that our method accurately selects the most relevant discriminative features.

The organization of the paper is as follows. The related work is delved into in Section 2. The proposed method is discussed in Section 3. In Section 4, experimental results are provided. Conclusion and future research are expressed in Section 5.

2. Related work

In this section, we describe the notations and definitions and provide a concise overview of two topics related to our work, i.e., SDA and semi-supervised sparse FS methods.

2.1. Notations and definitions

Suppose $X = [x_1, \ldots, x_n]^T \in \mathbb{R}^{d \times n}$ is the training data, and $X_l = [x_1, \ldots, x_l]^T \in \mathbb{R}^{d \times l}$ is the labeled training data, where l is the labeled data number and n is the total number of training data. $Y = [y_1, \ldots, y_n]^T \in \{0, 1\}^{n \times c}$ denotes the label of the training data, where $y_i \in \mathbb{R}^c$ $(1 \leq i \leq n)$ is the ith label vector and c is the number of classes. Y_{ij} is the jth label of y_i and $Y_{ij} = 1$ if x_i is in the jth class, while $Y_{ij} = 0$ otherwise. y_i is set to a vector with all zeros when x_i is not labeled. Some notations utilized in this paper are demonstrated in Table 1.

Semi-supervised sparse FS aims to apply all data and a sparse model to identify the most discriminative features. The sparse model ensures that the projection matrix exhibits sparsity in its rows, making it suitable for FS [26]. One of the sparse models widely used for semi-supervised FS is the $\ell_{2,1}$-norm. The $\ell_{2,p}$-norm $(0 < p \leq 1)$, defined by Eq. (1), extends the $\ell_{2,1}$-norm and is well-suited for semi-supervised

Table 1
Some notations utilized in this paper.

Notation	Description	Notation	Description
$X \in \mathbb{R}^{d\times n}$	The training data matrix	$\tilde{L}^t \in \mathbb{R}^{n\times n}$	The augmented matrix of L^t
$X_l \in \mathbb{R}^{d\times l}$	The labeled training data matrix	$H \in \mathbb{R}^{n\times n}$	The incidence matrix of a hypergraph
$Y \in \{0,1\}^{n\times c}$	The label matrix of training data	$d(v)$	The degree of the vertex v
$W \in \mathbb{R}^{d\times c}$	The projection matrix	$\delta(e)$	The degree of the hyperedge e
$S \in \mathbb{R}^{n\times n}$	The weight matrix of a simple graph	$A \in \mathbb{R}^{n\times n}$	The diagonal matrix of hyperedge weights
$L \in \mathbb{R}^{n\times n}$	The graph Laplacian matrix	$D^v \in \mathbb{R}^{n\times n}$	The diagonal matrix of the vertex degrees
$D \in \mathbb{R}^{n\times n}$	The diagonal matrix of $S, D_{ii} = \sum_j S_{ij}$	$D^e \in \mathbb{R}^{n\times n}$	The diagonal matrix of the hyperedge degrees
$S^w \in \mathbb{R}^{l\times l}$	The within-class graph similarity matrix for the labeled samples	$L_H \in \mathbb{R}^{n\times n}$	The hypergraph Laplacian matrix
$S^b \in \mathbb{R}^{l\times l}$	The between-class graph similarity matrix for the labeled samples	n	The number of training data
$L^w \in \mathbb{R}^{l\times l}$	The within-class Laplacian matrix	d	The number of features
$L^b \in \mathbb{R}^{l\times l}$	The between-class Laplacian matrix	l	The number of labeled data
$L^t \in \mathbb{R}^{l\times l}$	The total-class Laplacian matrix	u	The number of unlabeled data
$D^b \in \mathbb{R}^{l\times l}$	The diagonal matrix of $S^b, D^b_{ii} = \sum_j S^b_{ij}$	c	The number of classes
$D^w \in \mathbb{R}^{l\times l}$	The diagonal matrix of $S^w, D^w_{ii} = \sum_j S^w_{ij}$	$x_i \in \mathbb{R}^d$	The ith training sample
$\tilde{L}^b \in \mathbb{R}^{n\times n}$	The augmented matrix of L^b	$y_i \in \mathbb{R}^c$	The ith label vector

sparse FS. It provides enhanced sparsity and robustness compared to the $\ell_{2,1}$-norm.

$$\|W\|_{2,p} = \left(\sum_{i=1}^{d} \|w_i\|_2^p\right)^{1/p}, \quad p \in (0,1] \tag{1}$$

This model is converted to the convex $\ell_{2,1}$-norm if $p = 1$, and the non-convex norm if p belongs to $(0,1)$.

2.2. Brief review of SDA

LDA aims to determine a projection matrix for FE such that the samples of the same class are close to each other and those of various classes are far from each other. Suppose n_c is the number of samples in the cth class, $\mu^{(c)}$ the mean of the samples of the cth class, μ the mean of all samples, and x_i the ith sample. The between-class and total-scatter matrices in LDA are calculated as Eqs. (2) and (3), respectively:

$$M^b = \sum_{c=1}^{C} n_c(\mu^{(c)} - \mu)(\mu^{(c)} - \mu)^T \tag{2}$$

$$M^t = \sum_{i=1}^{n} (x_i - \mu)(x_i - \mu)^T \tag{3}$$

LDA can be optimized by the ratio trace criterion, which is equivalent to determinant ratio, as:

$$W^* = \underset{W}{\operatorname{argmax}} \operatorname{Tr}\left((W^T M^t W)^{-1}(W^T M^b W)\right) = \arg\max_W \frac{|W^T M^b W|}{|W^T M^t W|} \tag{4}$$

where W is the projection matrix utilized for FE.

Another optimization criterion for LDA can be achieved by trace ratio criterion as:

$$W^* = \underset{W}{\operatorname{argmax}} \frac{\operatorname{Tr}(W^T M^b W)}{\operatorname{Tr}(W^T M^t W)} = \underset{W}{\operatorname{argmin}} -\frac{\operatorname{Tr}(W^T M^b W)}{\operatorname{Tr}(W^T M^t W)} \tag{5}$$

Compared to the ratio trace problem, addressing the trace ratio problem often results in superior discriminative projections for various classification tasks [15]. However, it is challenging to obtain the closed-form solution of the trace ratio problem [27]. Consequently, many efforts focus on solving an equivalent trace difference problem instead of directly addressing the trace ratio problem. Let λ be the optimal trace ratio value satisfying

$$\lambda = \max_W \frac{\operatorname{Tr}(W^T M^b W)}{\operatorname{Tr}(W^T M^t W)} \tag{6}$$

As stated by Guo et al. [16], it follows that:

$$\max_W \operatorname{Tr}[W^T(M^b - \lambda M^t)W] = 0 \tag{7}$$

Therefore, drawing inspiration from Eq. (6), we present Theorem 1 from [16] without proof as follows:

Theorem 1. *The trace ratio problem can be equivalently solved by finding the zero point of the corresponding trace difference function defined as*

$$g(\lambda) = \max_W Tr[W^T(M^b - \lambda M^t)W] \tag{8}$$

By solving the trace difference equation $g(\lambda)$, the objective function in Eq. (5) can be transformed into Eq. (9). Then, the optimal projection matrix W^* can often be computed using an efficient iterative algorithm

$$W^* = \underset{W}{\operatorname{argmax}} \operatorname{Tr}[W^T(M^b - \lambda M^t)W] \tag{9}$$

Let $S^w \in \mathbb{R}^{l\times l}$ and $S^b \in \mathbb{R}^{l\times l}$ be two graph similarity matrices constructed by the labeled data and defined as:

$$S^w_{ij} = \begin{cases} \frac{1}{n_c} & \text{if } x_i \text{ and } x_j \text{ are both from the } c\text{th class} \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

$$S^b_{ij} = \begin{cases} \frac{1}{l} - \frac{1}{n_c} & \text{if } x_i \text{ and } x_j \text{ are both from the } c\text{th class} \\ \frac{1}{l} & \text{otherwise} \end{cases} \tag{11}$$

The graph Laplacian matrices are calculated as $L^w = D^w - S^w$ and $L^b = D^b - S^b$ such that $D^w_{ii} = \sum_j S^w_{ij}$ and $D^b_{ii} = \sum_j S^b_{ij}$. The within-class scatter matrix M^w, the between-class scatter matrix M^b, and the total scatter matrix M^t in LDA can be calculated as $M^w = X_l L^w X_l^T$, $M^b = X_l L^b X_l^T$, and $M^t = M^w + M^b = X_l L^t X_l^T$, where $L^t = L^w + L^b = I - \frac{1}{l}\mathbf{1}\mathbf{1}^T \in \mathbb{R}^{l\times l}$.

SDA [25] is a semi-supervised expansion of LDA, which employs a graph Laplacian constructed from both labeled and unlabeled data for FE. Its goal is to maximize the between-class distance while preserving the geometrical structure of the data through the use of a simple graph in the FE process. The objective function of SDA is defined as:

$$W^* = \underset{W}{\operatorname{argmax}} \frac{|W^T X \tilde{L}^b X^T W|}{|W^T(X(\tilde{L}^t + \alpha L)X^T + \beta I)W|} \tag{12}$$

where $\alpha \in \mathbb{R}^+$ and $\beta \in \mathbb{R}^+$ are the parameters that should be well-tuned. $L = D - S$ is the graph Laplacian where D is the diagonal matrix defined as $D_{ii} = \sum_j S_{ij}$ and S is the weight matrix of a graph can be formulated as

$$S_{ij} = \begin{cases} 1 & \text{if } x_i \in \text{KNN}(x_j) \text{ or } x_j \in \text{KNN}(x_i) \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

In Eq. (12), $\tilde{L}^b \in \mathbb{R}^{(l+u)\times(l+u)}$ and $\tilde{L}^t \in \mathbb{R}^{(l+u)\times(l+u)}$ are the augmented matrices of L^b and L^t, respectively, defined as follows:

$$\tilde{L}^b = \begin{bmatrix} L^b & 0 \\ 0 & 0 \end{bmatrix}, \quad \tilde{L}^t = \begin{bmatrix} L^t & 0 \\ 0 & 0 \end{bmatrix} \tag{14}$$

As previously noted, the determinant ratio problem is equivalent to the ratio trace one. Consequently, the objective function of SDA presented in Eq. (12) is in the ratio trace form, which can be solved by employing the generalized eigenvalue decomposition technique. For the eigen-problem to have a stable solution, the matrix $X(\tilde{L}^t + \alpha L)X^T$ in Eq. (12)

Table 2
Summary of the previous semi-supervised sparse FS methods and our proposed method.

Method	Sparse model used on regularization	Structural preservation	Complexity
SFSS [28]	$\ell_{2,1}$-norm	Simple graph	$O(n^3)$
FSLG [26]	$\ell_{2,1/2}$-norm	Simple graph	$O(n^3)$
SRS^{3}FS [22]	$\ell_{2,p}$-norm $(0 < p \leq 1)$	Spline regression	$O(n^3)$
Sr-SemiDFS [17]	$\ell_{1,2}$-norm	Simple graph	$O(d^3 + n^2) + O(n^3 + d^2)$
SFS [31]	$\ell_{2,1}$-norm	Simple graph	$O(d^3 + dn^2)$
ASLCGLFS [32]	$\ell_{2,1}$-norm	Adaptive simple graph	$\max(O(n^3), O(d^3))$
RDMRS2FS [30]	$\ell_{2,p}$-norm $(0 < p \leq 1)$	Simple dual graph	$\max(O(n^3), O(d^3))$
GS^{3}FS [29]	$\ell_{2,p}$-norm $(0 < p \leq 1)$	Simple graph	$\max(O(n^3), O(d^3))$
HSFSGU [33]	$\ell_{2,p}$-norm $(0 < p \leq 1)$	Hessian regularization	$O(d^3 + dn^2)$
HSDAFS (our)	$\ell_{2,p}$-norm $(0 < p \leq 1)$	Hypergraph	$O(d^3)$

must be non-singular. This condition is not met when there are more features than samples. For $\beta > 0$, the matrix $X(\tilde{L}^t + \alpha L)X^T + \beta I$ is certainly non-singular.

2.3. Review of several semi-supervised sparse FS methods

Recently, semi-supervised sparse FS methods have gained significant attention. Typically, the objective function of a semi-supervised sparse method integrates the manifold regularization, a loss function, and a sparse regularization term as depicted in Eq. (15), aimed at deriving the projection matrix W.

$$\underset{F,W,b}{\operatorname{argmin}} \operatorname{Tr}(F^T LF) + \operatorname{Tr}(F-Y)^T U(F-Y) + \alpha \operatorname{loss}(X^T W + \mathbf{1}_n b^T - F) + \beta R(W) \tag{15}$$

where $\operatorname{Tr}(\cdot)$ denotes the trace operator, $F = [f_1, f_2, \ldots, f_n]^T \in \mathbb{R}^{n\times c}$ is a predicted label matrix, $U \in \mathbb{R}^{n\times n}$ is a diagonal decision rule matrix in which $U_{ii} = \infty$ if x_i is labeled data and $U_{ii} = 1$ otherwise. $\operatorname{loss}(\cdot)$ represents the loss function, $\mathbf{1}_n \in \mathbb{R}^n$ is a vector of all 1, $b \in \mathbb{R}^c$ is the bias term, $R(W)$ denotes the regularization term, and α and β are the regularization parameters.

Various forms of loss function and regularization can be employed for different applications. The SFSS method [28] uses $\ell_{2,1}$-norm regularization to obtain the optimal projection matrix for FS. The FSLG method [26] applies $\ell_{2,1/2}$-norm regularization for selecting more sparse features. The SRS^{3}FS method [22] utilizes a spline regression matrix to maintain the geometrical structure of data and $\ell_{2,p}$-norm regularization to select the sparse features. Most semi-supervised sparse FS methods [26,28] utilize the $\|X^T W + \mathbf{1}_n b^T - F\|_F^2$ square norm-based loss function that is sensitive to outliers. The GS^{3}FS method [29] uses the $\ell_{2,p}$-norm on both loss function and regularization that is robust to outliers and selects the sparse features. The RDMRS2FS method [30] is an FS method based on minimum redundancy through self-representation and robust dual-graph regularization. It integrates self-representation within a sparse learning framework, utilizing $\ell_{2,p}$-norm minimization in both regularization and loss functions. We provide a summary of several semi-supervised sparse FS methods in Table 2 from various perspectives, including the preservation of structured information, the sparse model used for regularization, and the computational complexity of the optimization algorithm.

Compared with the other FS methods outlined in Table 2, the proposed HSDAFS method preserves the structure of data through hypergraph Laplacian, enabling it to capture complex and high-order relationships among data. Additionally, the use of $\ell_{2,p}$-norm regularization in the proposed method results in the selection of more sparse features. Moreover, the HSDAFS method leverages the discriminative information of labeled data, giving it an advantage over traditional regression-based models due to its superior discriminative capability.

3. Proposed method

In this section, we propose a sparse FS method using the hypergraph Laplacian-based SDA, called HSDAFS, and present an algorithm to solve the proposed HSDAFS. This section involves describing hypergraph learning, proposing framework of the HSDAFS method, presenting an algorithm to solve it, and conducting convergence analysis of the algorithm.

3.1. Hypergraph learning

Recent researches in graph learning have shown the widespread use of simple graphs to preserve geometrical structures in data, relying on pairwise relationships between data points. However, real-world data relationships are often more complex than the simple pairwise relationships. Representing such complexity with pairwise relationships results in a significant loss of intrinsic structural information [24,34]. To overcome this limitation, hypergraphs have emerged as a crucial tool representing the intricate high-order structure of data. Unlike simple graphs, which model relationships in a pairwise manner, hypergraphs can represent higher-order relationships. Each hyperedge in a hypergraph can connect multiple vertices simultaneously, allowing for the capture of more complex interactions [35].

Constructing a hypergraph, denoted as $G = (V, E, A)$, begins with defining the sets of vertices and hyperedges, represented by V and E, respectively. Each hyperedge $e \in E$ represents an arbitrary subset of V, grouping data samples based on specific relations. The weights of these hyperedges are represented by the diagonal matrix A, with each diagonal element $a(e)$ corresponding to the weight of hyperedge e. The next step involves representing the structure of the hypergraph by creating the incidence matrix H, which encodes the relationships between vertices and hyperedges. The incidence matrix is defined as:

$$H(v,e) = \begin{cases} 1 & \text{if } v \in e \\ 0 & \text{if } v \notin e \end{cases} \tag{16}$$

where v denotes a vertex belonging to V.

The incidence matrix captures the vertex-to-hyperedge relationships rather than vertex-to-vertex relationships, enabling the representation of higher-order relationships. To obtain a vertex-to-vertex relationship, which is necessary for many computations, the Laplacian matrix of the hypergraph should be constructed. One possible method to achieve this is by defining the unnormalized Laplacian matrix of a hypergraph based on the incidence matrix as follows:

$$L_H = D^v - HA(D^e)^{-1}H^T \tag{17}$$

where D^v and D^e are diagonal matrices representing the degrees of the vertices and hyperedges, respectively. The vertex degree $d(v)$, calculated as Eq. (18), is the sum of the weights of the hyperedges connected to it, and the hyperedge degree $\delta(e)$, calculated as Eq. (19), is the sum of the incidences of the vertices within it.

$$d(v) = \sum_{e\in E} a(e) \cdot H(v,e) \tag{18}$$

$$\delta(e) = \sum_{v \in V} H(v, e) \tag{19}$$

The formulation of Eq. (17) allows the hypergraph Laplacian to encapsulate the complex relationships between multiple vertices connected by hyperedges, which a simple graph cannot achieve. By leveraging this sophisticated structure, the hypergraph Laplacian improves the ability to capture the intrinsic geometrical structure of the data, leading to enhanced performance in tasks such as feature selection and discriminant analysis.

It is worth mentioning that hyperedge construction plays an important role in hypergraph learning for extracting high-order structural information from data. In this paper, we employ a K-nearest neighbors (KNN) technique for constructing hypergraphs. Specifically, each sample is treated as a vertex, and the goal is to generate a hyperedge for each vertex. To achieve this, we compute the K-nearest neighbors for each vertex using the Euclidean distance between the vertex and other vertices. Consequently, a hyperedge is formed for each vertex by connecting it with its K-nearest neighbors. With n samples, we construct n hyperedges. Following [36], we assign equal weighting values to all hyperedges, setting $a(e) = 1$ for the hyperedge e.

3.2. Framework of the proposed HSDAFS method

Discriminant analysis plays an effective role in FS by identifying the most discriminative features. LDA model is particularly advantageous over traditional regression models due to its superior discriminative capability. Furthermore, in many real-world applications, there exists a limited amount of labeled data and a substantial volume of unlabeled data, which can be effectively utilized in semi-supervised FS. SDA [25] is a semi-supervised FE technique that leverages both the discriminative information from labeled data and the manifold structure of all data based on a simple graph to create a projection matrix to transform the original feature space into a low-dimensional one. However, the similarity matrix created by a simple graph may not capture the complex relationships among data points. Hypergraphs may capture more complex data relationships compared to simple graphs while preserving the local structure of the data [34]. Moreover, as mentioned earlier, the objective function of SDA is in the ratio trace form. Empirical evidence shows that trace ratio-based LDA often outperforms ratio trace-based LDA in many classification tasks.

In light of the aforementioned issues, we propose a semi-supervised FS method based on SDA that leverages the discriminative information of labeled data and the geometrical structure of all data using hypergraph Laplacian. This method reformulates the objective function of SDA into a trace ratio form for an FS framework, aiming to select the most discriminative features. Inspired by the SDA-based FE framework in Eq. (12), our proposed method takes the trace ratio form for FS, employing hypergraph Laplacian and the mixed $\ell_{2,p}$-norm regularization. We utilize the k-nearest neighbor strategy to construct a hypergraph and subsequently derive a Laplacian matrix from this hypergraph. The hypergraph in our method differs significantly from a simple KNN graph. Unlike the KNN graph, our hypergraph can capture higher-order relationships where a single hyperedge connects multiple vertices simultaneously, facilitating more complex interactions beyond the pairwise connections seen in KNN graphs. This higher-order connectivity allows the hypergraph Laplacian to better preserve the intrinsic geometrical structure of the data, thereby enhancing its ability to capture the underlying data manifold. Leveraging these higher-order relationships enhances the discriminative power of our model and leads to improved performance.

In the proposed method, we aim to calculate a projection matrix $W \in \mathbb{R}^{d \times c}$ for FS that keeps the discriminative information and the local structure of data using a hypergraph Laplacian. The $\ell_{2,p}$-norm regularization shrinks some rows of W to zero, making the matrix appropriate for FS. Therefore, we propose our hypergraph Laplacian-based semi-supervised discriminant analysis feature selection (HSDAFS) method using the trace ratio criterion as follows:

$$W^* = \underset{W}{\operatorname{argmin}} -\frac{\operatorname{Tr}(W^T X \tilde{L}^b X^T W)}{\operatorname{Tr}(W^T (X(\tilde{L}^t + \alpha(D^v - HAD^{e-1}H^T))X^T + \beta I)W)} + \gamma \|W\|_{2,p}^p, \quad p \in (0, 1] \tag{20}$$

where $L_H = D^v - HAD^{e-1}H^T$ is the hypergraph Laplacian matrix and γ is the regularization parameter applied for controlling the regularization effect.

As stated in Section 2.2, it is difficult to obtain the closed-form solution for the trace ratio problem [27]. Consequently, we transform the proposed trace ratio-based HSDAFS method in Eq. (20) into an equivalent trace difference method according to Theorem 1, expressed as follows:

$$W^* = \underset{W}{\operatorname{argmin}} -\operatorname{Tr}[W^T (X \tilde{L}^b X^T - \lambda(X(\tilde{L}^t + \alpha(D^v - HAD^{e-1}H^T))X^T + \beta I))W] + \gamma \|W\|_{2,p}^p \tag{21}$$

Our FS method reformulates the FE-based SDA framework into a trace ratio form and transforms it into an FS-based framework. It exploits the discriminative ability of SDA, which leads to its superiority over traditional regression-based semi-supervised FS methods. By incorporating semi-supervised discriminant analysis, hypergraph Laplacian, and $\ell_{2,p}$-norm regularization, our method maximizes the between-class distance, preserves the geometrical structure of all data, maintains high-order relationships in the data and considers the correlation among various features in the FS process. This method ensures the effective selection of discriminative features, leading to improved learning performance.

3.3. Optimization and algorithm

The formulation in Eq. (20) comprises the mixed $\ell_{2,p}$-norm, which is non-smooth and its optimization is complex because we should solve the geometrically-based regularized LDA problem to acquire the sparse projection matrix W for FS. Moreover, the $\ell_{2,p}$-norm based method exhibits convexity when $p = 1$ and non-convexity when p belongs to $(0, 1)$. Accordingly, we propose to obtain the solution for the objective function as follows:

Given $W = [w_1, \ldots, w_d]^T$, a diagonal matrix D is defined whose diagonal elements are $D_{ii} = \frac{p}{2\|w_i\|_2^{2-p}}$. According to the definition of $\ell_{2,p}$-norm, $\|W\|_{2,p}^p = \frac{2}{p}\operatorname{Tr}(W^T DW)$. Therefore, Eq. (20) can be equivalently expressed as:

$$\mathcal{J}(W) = \underset{W}{\operatorname{argmin}} -\frac{\operatorname{Tr}(W^T X \tilde{L}^b X^T W)}{\operatorname{Tr}(W^T (X(\tilde{L}^t + \alpha L_H)X^T + \beta I)W)} + \gamma \frac{2}{p}\operatorname{Tr}(W^T DW) \tag{22}$$

Since Eq. (22) is a trace ratio optimization problem, obtaining the closed-form solution is challenging. Therefore, based on Theorem 1, the trace ratio method in Eq. (22) is converted into the trace difference method as:

$$\mathcal{J}(W) = \underset{W}{\operatorname{argmin}} -\operatorname{Tr}\left[W^T (X \tilde{L}^b X^T - \lambda(X(\tilde{L}^t + \alpha L_H)X^T + \beta I))W\right] + \gamma \frac{2}{p}\operatorname{Tr}(W^T DW) \tag{23}$$

Simplifying further, we get:

$$\mathcal{J}(W) = \underset{W}{\operatorname{argmin}} -\operatorname{Tr}(W^T X \tilde{L}^b X^T W) + \lambda \operatorname{Tr}(W^T X \tilde{L}^t X^T W) + \lambda\alpha \operatorname{Tr}(W^T X L_H X^T W) + \lambda\beta \operatorname{Tr}(W^T W) + \gamma \frac{2}{p}\operatorname{Tr}(W^T DW) \tag{24}$$

By taking the derivative of Eq. (24) with respect to W and setting it to zero to find the optimal W, we have:

$$\frac{\partial \mathcal{J}(W)}{\partial W} = -2X \tilde{L}^b X^T W + 2\lambda X \tilde{L}^t X^T W + 2\lambda\alpha X L_H X^T W + 2\lambda\beta W$$

Algorithm 1: The proposed unified iterative algorithm to solve (20).

Input : Data matrix $X \in \mathbb{R}^{d\times n}$;
Label matrix $Y \in \mathbb{R}^{n\times c}$;
Regularization parameters α, β and γ;
Output: The index of top k selected features;
begin
1. Construct the hypergraph Laplacian matrix $L_H \in \mathbb{R}^{n\times n}$;
2. Construct the matrix $\tilde{L}^t \in \mathbb{R}^{n\times n}$ as the augmented matrix of L^t;
3. Construct the matrix $\tilde{L}^b \in \mathbb{R}^{n\times n}$ as the augmented matrix of L^b;
4. Set $t = 0$ and initialize $W^t \in \mathbb{R}^{d\times c}$ randomly;
5. **Repeat**
Calculate $W = [w_1^t, \dots, w_d^t]^T$;
Compute the diagonal matrix D^t according to $D_{ii}^t = \frac{p}{2\|w_i^t\|_2^{2-p}}$;
$U^t = \left(X(\tilde{L}^t + \alpha L_H)X^T + \beta I\right)^{-1}\left(XL\tilde{L}^bX^T - \gamma\frac{2}{p}D^t\right)$;
Update $W^{(t+1)}$ as $W^{(t+1)} = [u_1, u_2, \dots, u_q]$, where $u_1, u_2, \dots, u_q$ are the eigenvectors of U^t corresponding to the q smallest eigenvalues;
$t = t + 1$;
Until Convergence;
6. Sort each feature in accordance with $\|w_i\|_2$ in non-increasing order and choose the top k ranked features;

$$+ 2\gamma\frac{2}{p}DW = 0 \tag{25}$$

Simplifying, we obtain:

$$(X\tilde{L}^bX^T - \gamma\frac{2}{p}D)W = \lambda(X(\tilde{L}^t + \alpha L_H)X^T + \beta I)W \tag{26}$$

which can be rewritten as a generalized eigenvalue problem:

$$(X(\tilde{L}^t + \alpha L_H)X^T + \beta I)^{-1}(X\tilde{L}^bX^T - \gamma\frac{2}{p}D)W = \lambda W \tag{27}$$

Eq. (27) represents an eigenvalue problem. To obtain W for FS, we conduct an eigen-decomposition of $(X(\tilde{L}^t + \alpha L_H)X^T + \beta I)^{-1}(X\tilde{L}^bX^T - \gamma\frac{2}{p}D)$, where W is the projection matrix used for FS.

We propose Algorithm 1 to obtain the solution for our proposed formulation in Eq. (20) for all $p \in (0, 1]$, and verify its convergence using Theorem 2.

In Step 5 of Algorithm 1, we propose an iterative approach to optimize the minimization problem in (20). The iterative approach allows for updating the projection matrix W iteratively to find the optimal FS. This process ensures that the value of the objective function decreases monotonically with each iteration until it converges to the optimal W. In Section 3.4, we prove in Theorem 2 that the proposed iterative approach in Algorithm 1 converges to the optimal W corresponding to (20). This convergence proof guarantees that the iterative process is not only effective but also reliable for achieving the optimal solution.

Here we briefly discuss the computational complexity of Algorithm 1. The complexity of the calculation of the hypergraph Laplacian matrix and the augmented matrices of L^t and L^b is $O(n^2)$. The time complexity for solving the eigen problem $(X(\tilde{L}^t + \alpha L_H)X^T + \beta I)^{-1}(X\tilde{L}^bX^T - \gamma\frac{2}{p}D)W = \lambda W$ is $O(d^3)$. The time complexity for sorting the features is $O(d \log d)$.

3.4. Convergence analysis

We prove the convergence analysis of Algorithm 1 utilizing Theorem 2. Two following lemmas introduced in [18] are needed to prove Theorem 2.

Lemma 1. *If* $\phi(t) = \frac{2}{2-p}t^{-\frac{p}{2-p}}$ *when* $p \in (0, 1)$ *and* $t > 0$, *therefore,* $\phi(t) \le 0$.

Lemma 2. *Assume* g_i^t *and* g_i^{t+1} *as the* i*th row of matrices* G_t *and* G_{t+1} *respectively, thus for* $p \in (0, 1]$ *we have:*

$$\left\|g_i^{t+1}\right\|_2^p - \frac{p}{2}\left(\frac{\left\|g_i^{t+1}\right\|_2^2}{\left\|g_i^t\right\|_2^{2-p}}\right) \le \left\|g_i^t\right\|_2^p - \frac{p}{2}\left(\frac{\left\|g_i^t\right\|_2^2}{\left\|g_i^t\right\|_2^{2-p}}\right)$$

$$\Longrightarrow \sum_{i=1}^{r}\left(\left\|g_i^{t+1}\right\|_2^p - \frac{p}{2}\left(\frac{\left\|g_i^{t+1}\right\|_2^2}{\left\|g_i^t\right\|_2^{2-p}}\right)\right) \le \sum_{i=1}^{r}\left(\left\|g_i^t\right\|_2^p - \frac{p}{2}\left(\frac{\left\|g_i^t\right\|_2^2}{\left\|g_i^t\right\|_2^{2-p}}\right)\right)$$

Theorem 2. *For* $0 < p \le 1$, *the iterative approach in Algorithm 1 monotonically decreases the value of the objective function in Eq. (20) in each iteration until it converges to the optimal* W.

Proof. We deduce the following formula from Eq. (23):

$$W^{(t+1)} = \underset{W}{\operatorname{argmin}}\left\{-\mathrm{Tr}\left(W^T(X\tilde{L}^bX^T - \lambda X(\tilde{L}^t + \alpha L_H)X^T - \lambda\beta I)W\right) + \gamma\mathrm{Tr}(W^TD^tW)\right\}$$

Hence, we have:

$$-\mathrm{Tr}\left(W^{(t+1)^T}(X\tilde{L}^bX^T - \lambda X(\tilde{L}^t + \alpha L_H)X^T - \lambda\beta I)W^{(t+1)}\right) + \gamma\mathrm{Tr}(W^{(t+1)^T}D^tW^{(t+1)}) \le -\mathrm{Tr}\left(W^{t^T}(X\tilde{L}^bX^T - \lambda X(\tilde{L}^t + \alpha L_H)X^T - \lambda\beta I)W^t\right) + \gamma\mathrm{Tr}(W^{t^T}D^tW^t)$$

$$\mathrm{Tr}(W^tD^tW^t) = \sum_{i=1}^{d}\frac{p}{2}\left(\frac{\|w_i^t\|_2^2}{\|w_i^t\|_2^{2-p}}\right), \quad \mathrm{Tr}(W^{(t+1)^T}D^tW^{(t+1)}) = \sum_{i=1}^{d}\frac{p}{2}\left(\frac{\left\|w_i^{(t+1)}\right\|_2^2}{\|w_i^t\|_2^{2-p}}\right)$$

$$\begin{aligned}
&\Longrightarrow -\mathrm{Tr}\left(W^{(t+1)^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^{(t+1)}\right)\\
&\quad+\gamma\sum_{i=1}^{d}\frac{p}{2}\left(\frac{\left\|w_i^{(t+1)}\right\|_2^2}{\left\|w_i^t\right\|_2^{2-p}}\right)\\
&\leq -\mathrm{Tr}\left(W^{t^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^t\right)\\
&\quad+\gamma\sum_{i=1}^{d}\frac{p}{2}\left(\frac{\left\|w_i^t\right\|_2^2}{\left\|w_i^t\right\|_2^{2-p}}\right)\\
&\Longrightarrow -\mathrm{Tr}\left(W^{(t+1)^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^{(t+1)}\right)\\
&\quad+\gamma\sum_{i=1}^{d}\frac{p}{2}\left(\frac{\left\|w_i^{(t+1)}\right\|_2^2}{\left\|w_i^t\right\|_2^{2-p}}\right)-\gamma\sum_{i=1}^{d}\left\|w_i^{(t+1)}\right\|_2^p+\gamma\sum_{i=1}^{d}\left\|w_i^{(t+1)}\right\|_2^p\\
&\leq -\mathrm{Tr}\left(W^{t^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^t\right)\\
&\quad+\gamma\sum_{i=1}^{d}\frac{p}{2}\left(\frac{\left\|w_i^t\right\|_2^2}{\left\|w_i^t\right\|_2^{2-p}}\right)-\gamma\sum_{i=1}^{d}\|w_i^t\|_2^p+\gamma\sum_{i=1}^{d}\|w_i^t\|_2^p\\
&\Longrightarrow -\mathrm{Tr}\left(W^{(t+1)^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^{(t+1)}\right)\\
&\quad+\gamma\sum_{i=1}^{d}\left\|w_i^{(t+1)}\right\|_2^p-\gamma\left(\sum_{i=1}^{d}\left\|w_i^{(t+1)}\right\|_2^p-\sum_{i=1}^{d}\frac{p}{2}\left(\frac{\left\|w_i^{(t+1)}\right\|_2^2}{\left\|w_i^t\right\|_2^{2-p}}\right)\right)\\
&\leq -\mathrm{Tr}\left(W^{t^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^t\right)\\
&\quad+\gamma\sum_{i=1}^{d}\|w_i^t\|_2^p-\gamma\left(\sum_{i=1}^{d}\|w_i^t\|_2^p-\sum_{i=1}^{d}\frac{p}{2}\left(\frac{\left\|w_i^t\right\|_2^2}{\left\|w_i^t\right\|_2^{2-p}}\right)\right)
\end{aligned}$$

It is proved in Lemma 2 that

$$\sum_{i=1}^{r}\left(\|w_i^{(t+1)}\|_2^p-\frac{p}{2}\frac{\|w_i^{(t+1)}\|_2^2}{\|w_i^t\|_2^{2-p}}\right)\leq\sum_{i=1}^{r}\left(\|w_i^t\|_2^p-\frac{p}{2}\frac{\|w_i^t\|_2^2}{\|w_i^t\|_2^{2-p}}\right)$$

Thus, we have

$$\begin{aligned}
&-\mathrm{Tr}\left(W^{(t+1)^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^{(t+1)}\right)\\
&\quad+\gamma\sum_{i=1}^{d}\|w_i^{(t+1)}\|_2^p\\
&\leq -\mathrm{Tr}\left(W^{t^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^t\right)+\gamma\sum_{i=1}^{d}\|w_i^t\|_2^p\\
&\quad\Longrightarrow -\mathrm{Tr}\left(W^{(t+1)^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^{(t+1)}\right)\\
&\quad+\gamma\|w_i^{(t+1)}\|_{2,p^p}\\
&\leq -\mathrm{Tr}\left(W^{t^T}\left(X\tilde{L}^bX^T-\lambda X(\tilde{L}^t+\alpha L_H)X^T-\lambda\beta I\right)W^t\right)+\gamma\|w_i^t\|_{2,p}^p
\end{aligned}$$

which proves the correctness of Theorem 2. □

4. Experiment

To evaluate the performance of our HSDAFS method in selecting discriminative features, we conduct several experiments on six publicly available datasets, as presented in Table 3. The source code of the HSDAFS method and the datasets used are available on GitHub (https://github.com/rsheikhpour/SDAFS).

4.1. Compared methods

The proposed HSDAFS method is compared to eight semi-supervised sparse FS methods with the same experimental setup. A brief description of each method is provided below, and a summary is presented in Table 2.

Table 3
Details of the datasets.

Dataset	#Size	#Features	#Classes	Type	Percentage of labeled data
COIL20	1440	1024	20	Image	10%, 25%, 50%
PIE	3332	1024	68	Image	10%, 25%, 50%
UMIST	575	1024	20	Image	10%, 25%, 50%
WarpPIE10P	210	2420	10	Image	10%, 25%, 50%
WrapAR10P	130	2400	10	Image	10%, 25%, 50%
ISOLET	1560	617	26	Voice	10%, 25%, 50%

- SFSS [28]: Utilizes manifold regularization based SSL, with the loss function based on the square norm and $\ell_{2,1}$-based regularization to compute the sparse projection matrix for FS.
- FSLG [26]: Employs SSL based on manifold regularization, with the loss function based on the square norm and $\ell_{2,1/2}$-based regularization to consider the correlation between features and jointly select them.
- SRS^{3}FS [22]: Utilizes SSL based on spline regression to capture the local structure of data, with the square norm-based loss function and the mixed $\ell_{2,p}$-norm ($0<p\leq 1$)-based regularization to jointly select relevant features from all data. In our experiments, the SRS^{3}FS method is used based on $\ell_{2,1/2}$-norm and $\ell_{2,1}$-norm.
- Sr-SemiDFS [17]: A structure regularized semi-supervised discriminative FS framework that combines data structure learning with LDA-based FS into a single framework and learns soft and hard data structures.
- SFS [31]: Employs a generalized uncorrelated constraint and manifold embedding using graph Laplacian to identify uncorrelated and discriminative features.
- ASLCGLFS [31]: Integrates adaptive structure learning and constrained graph learning into a framework, considering both local and global manifold structures simultaneously. It expands adaptive graph learning by incorporating label information, aiming to enhance the quality of the similarity matrix through constrained graph learning.
- RDMRS2FS [30]: Adopts minimum redundancy based self-representation and robust dual-graph regularization within the semi-supervised FS framework. It operates as a dual-graph regularized FS method, constructing data and feature graphs to capture the manifold structural information from both data and feature spaces.

4.2. Experimental setting

In the experiments, SVM, KNN (1-Nearest Neighbor), and Random Forest (RF) classifiers are applied to evaluate the performance of HSDAFS and other FS methods. We set or tune the parameter values applied in the experiments based on the values reported in the previous literature [22,28]. In the experiments, we set the parameter k to 5, representing the number of nearest neighbors used for calculating both graph Laplacian and hypergraph Laplacian. Parameter p in our method is implemented from $\{0.25, 0.5, 0.75, 1\}$. All datasets are randomly split into training and test sets with the ratio of 70% and 30%, respectively. The training set is also split into labeled (10%, 25%, 50% of training samples) and unlabeled sets (90%, 75%, 50% of training samples). This process is performed three times using various percentages of labeled data; the experiments are repeated on various generated datasets, and the mean accuracies are reported. We select the top 50 ranked features for the ISOLET dataset and the top 100 ranked features for the other datasets. We tune the regularization parameters α, β, and γ in Eq. (20) and other FS methods from $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$ utilizing a grid-search.

Table 4
Classification accuracy (Mean ± Std) of SVM classifier using different FS methods on different datasets.

L%	Method	COIL20	PIE	UMIST	WarpPIE10P	WarpAR10P	ISOLET
10%	FSLG	0.6829 ± 0.0267	0.2733 ± 0.0469	0.5106 ± 0.0297	0.4233 ± 0.0183	0.3163 ± 0.0592	0.5599 ± 0.0060
	SFSS	0.7176 ± 0.0324	0.4223 ± 0.0281	0.4778 ± 0.0297	0.4286 ± 0.0275	0.2564 ± 0.0678	0.5224 ± 0.0287
	SRS^{3}FS ($\ell_{2,1/2}$)	0.7114 ± 0.0337	0.3710 ± 0.003	0.5318 ± 0.0208	0.4709 ± 0.0367	0.3077 ± 0.1175	0.5599 ± 0.0030
	SRS^{3}FS ($\ell_{2,1}$)	0.7369 ± 0.0368	0.4190 ± 0.0418	0.5125 ± 0.0665	0.4497 ± 0.0510	0.2735 ± 0.1036	0.5331 ± 0.0136
	Sr-SemiDFS	0.7114 ± 0.0372	0.2150 ± 0.0210	0.4971 ± 0.0252	0.4286 ± 0.0965	0.1966 ± 0.0784	0.4167 ± 0.0030
	SFS	0.7169 ± 0.0415	0.4170 ± 0.0408	0.5106 ± 0.0349	0.4762 ± 0.0476	0.2649 ± 0.0645	0.5310 ± 0.0136
	ASLCGLFS	0.7060 ± 0.0401	0.2713 ± 0.0546	0.5318 ± 0.0306	0.4549 ± 0.0331	0.2051 ± 0.0513	0.4188 ± 0.0604
	RDMRS2FS	0.6821 ± 0.0756	0.3510 ± 0.0322	0.5395 ± 0.0133	0.4762 ± 0.0420	0.2137 ± 0.0592	0.4883 ± 0.0076
	HSDAFS ($\ell_{2,1/4}$)	**0.7438 ± 0.0197**	0.4130 ± 0.0322	0.5144 ± 0.0209	**0.5449 ± 0.0641**	**0.3334 ± 0.0444**	0.5649 ± 0.0141
	HSDAFS ($\ell_{2,1/2}$)	0.7315 ± 0.0209	0.4257 ± 0.0299	0.5433 ± 0.0514	0.5291 ± 0.0485	0.3077 ± 0.0444	**0.5727 ± 0.0121**
	HSDAFS ($\ell_{2,3/4}$)	0.7400 ± 0.0227	**0.4317 ± 0.0268**	**0.5626 ± 0.0033**	0.5185 ± 0.0367	0.2650 ± 0.0148	0.5396 ± 0.0227
	HSDAFS ($\ell_{2,1}$)	0.7153 ± 0.0167	0.4197 ± 0.0182	0.5395 ± 0.0033	0.5397 ± 0.0420	0.3254 ± 0.0544	0.5054 ± 0.0076
25%	FSLG	0.8557 ± 0.0243	0.5727 ± 0.1050	0.8376 ± 0.0358	0.6720 ± 0.0814	0.4957 ± 0.0296	0.7161 ± 0.0086
	SFSS	0.8534 ± 0.0232	0.5913 ± 0.0707	0.8381 ± 0.0361	0.7196 ± 0.0331	0.4530 ± 0.0592	0.7158 ± 0.0242
	SRS^{3}FS ($\ell_{2,1/2}$)	0.8603 ± 0.0334	0.6163 ± 0.0519	0.8402 ± 0.0385	0.7301 ± 0.0420	0.4957 ± 0.0392	0.7212 ± 0.0076
	SRS^{3}FS ($\ell_{2,1}$)	0.8601 ± 0.0228	0.6133 ± 0.0881	0.8478 ± 0.0291	0.7249 ± 0.0242	0.4530 ± 0.0592	0.7180 ± 0.0211
	Sr-SemiDFS	0.7639 ± 0.0093	0.2653 ± 0.0068	0.8613 ± 0.0058	0.6667 ± 0.0420	0.3504 ± 0.0148	0.5534 ± 0.0513
	SFS	0.8503 ± 0.0048	0.5703 ± 0.0283	0.8266 ± 0.0200	0.7249 ± 0.0183	0.4701 ± 0.0148	0.7148 ± 0.0227
	ASLCGLFS	0.8087 ± 0.0128	0.5640 ± 0.0635	0.8304 ± 0.0120	0.6825 ± 0.0692	0.3932 ± 0.0148	0.6037 ± 0.0499
	RDMRS2FS	0.8379 ± 0.0212	0.5007 ± 0.0193	0.8536 ± 0.0372	0.7037 ± 0.0458	0.4162 ± 0.0503	0.7083 ± 0.0136
	HSDAFS ($\ell_{2,1/4}$)	**0.8727 ± 0.0092**	0.6637 ± 0.0307	0.8748 ± 0.0177	**0.7513 ± 0.0242**	**0.5556 ± 0.0645**	0.7500 ± 0.0030
	HSDAFS ($\ell_{2,1/2}$)	0.8719 ± 0.0093	**0.6657 ± 0.0193**	**0.8882 ± 0.0120**	0.7143 ± 0.0420	0.5470 ± 0.0901	**0.7682 ± 0.0015**
	HSDAFS ($\ell_{2,3/4}$)	0.8603 ± 0.0237	0.6603 ± 0.0287	0.8632 ± 0.0376	0.6931 ± 0.0661	0.5470 ± 0.0148	0.7479 ± 0.0211
	HSDAFS ($\ell_{2,1}$)	0.8549 ± 0.0256	0.6567 ± 0.0226	0.8806 ± 0.0464	0.7460 ± 0.0159	0.5043 ± 0.0534	0.7479 ± 0.0182
50%	FSLG	0.9051 ± 0.0379	0.6033 ± 0.0214	0.8632 ± 0.0273	0.9471 ± 0.0184	0.6923 ± 0.1427	0.8451 ± 0.0045
	SFSS	0.9182 ± 0.0256	0.7187 ± 0.1197	0.8902 ± 0.0379	0.9577 ± 0.0242	0.7094 ± 0.1156	0.8280 ± 0.0106
	SRS^{3}FS ($\ell_{2,1/2}$)	0.9368 ± 0.0194	0.7023 ± 0.0745	0.8960 ± 0.0321	0.9577 ± 0.0184	0.7179 ± 0.0770	0.8334 ± 0.0121
	SRS^{3}FS ($\ell_{2,1}$)	0.9367 ± 0.0136	0.7837 ± 0.0124	0.8960 ± 0.0289	0.9471 ± 0.0330	0.7265 ± 0.0901	0.8366 ± 0.0015
	Sr-SemiDFS	0.8325 ± 0.0578	0.2933 ± 0.0426	0.9210 ± 0.0186	0.9312 ± 0.0242	0.5983 ± 0.0784	0.6624 ± 0.0303
	SFS	0.8388 ± 0.161	0.6823 ± 0.0394	0.9268 ± 0.0234	0.9630 ± 0.0092	0.6538 ± 0.1000	0.8408 ± 0.0317
	ASLCGLFS	0.9097 ± 0.0140	0.0.491 ± 0.0590	0.9210 ± 0.0186	0.9683 ± 0.0159	0.6667 ± 0.1428	0.6485 ± 0.0590
	RDMRS2FS	0.9290 ± 0.0109	0.7730 ± 0.0933	0.9231 ± 0.0316	0.9682 ± 0.0275	0.6752 ± 0.0971	0.8398 ± 0.0030
	HSDAFS ($\ell_{2,1/4}$)	0.9398 ± 0.0223	**0.8110 ± 0.0234**	0.9306 ± 0.0209	0.9736 ± 0.0242	**0.7436 ± 0.0444**	**0.8782 ± 0.0151**
	HSDAFS ($\ell_{2,1/2}$)	0.9429 ± 0.0140	0.7960 ± 0.0240	0.9152 ± 0.0203	**0.9841 ± 0.0159**	0.7351 ± 0.0645	0.8600 ± 0.0137
	HSDAFS ($\ell_{2,3/4}$)	**0.9514 ± 0.0140**	0.7873 ± 0.0204	0.9307 ± 0.0300	0.9735 ± 0.0183	0.6838 ± 0.0783	0.8590 ± 0.0091
	HSDAFS ($\ell_{2,1}$)	0.9383 ± 0.0197	0.8027 ± 0.0280	**0.9383 ± 0.0329**	0.9788 ± 0.0091	0.7094 ± 0.0971	0.8526 ± 0.0000

4.3. Experimental results

The results of the SVM classifier obtained using various FS methods on all datasets are provided in Table 4.

As indicated in Table 4, when utilizing 10% of labeled data, the highest accuracy of SVM on the COIL20, WarpPIE10P, and WarpAR10P datasets is achieved using HSDAFS based on $\ell_{2,1/4}$-norm. On the PIE and UMIST datasets, HSDAFS based on $\ell_{2,3/4}$-norm yields the best results, and on the ISOLET dataset, HSDAFS based on $\ell_{2,1/2}$-norm achieves the highest accuracy. When the percentage of labeled data is 25%, HSDAFS based on $\ell_{2,1/4}$-norm demonstrates superior performance compared to other methods on the COIL20, WarpPIE10P, and WarpAR10P datasets. On the PIE, UMIST, and ISOLET datasets, HSDAFS based on $\ell_{2,1/2}$-norm outperforms other methods. With 50% of labeled data, the proposed HSDAFS based on $\ell_{2,1/4}$-norm achieves the best results on the PIE, WarpAR10P, and ISOLET datasets, while on the COIL20 dataset, HSDAFS based on $\ell_{2,3/4}$-norm performs the best. Additionally, HSDAFS based on $\ell_{2,1}$-norm and HSDAFS based on $\ell_{2,1/2}$-norm achieve the highest accuracy on the UMIST and WarpPIE10P datasets, respectively.

The results of KNN classifier using various FS methods on all datasets are presented in Table 5.

From Table 5, it is evident that the KNN classifier achieves its best accuracy with 10% of labeled data on the COIL20 dataset using HSDAFS based on the $\ell_{2,3/4}$-norm, and on the PIE and ISOLET datasets using HSDAFS based on $\ell_{2,1/2}$-norm. Similarly, the proposed HSDAFS based on $\ell_{2,1/4}$-norm performs the best on the UMIST and WarpAR10P datasets. On the WarpPIE10P dataset, the highest accuracy is attained by HSDAFS based on $\ell_{2,1/2}$-norm and $\ell_{2,3/4}$-norm. When 25% of data are labeled, the proposed HSDAFS based on the $\ell_{2,1/2}$-norm outperforms other FS methods on the COIL20, PIE, and WarpAR10P datasets. HSDAFS based on the $\ell_{2,1}$-norm shows superior performance on the WarpPIE10P and ISOLET datasets. Moreover, the highest accuracy on the UMIST dataset is achieved by HSDAFS based on the $\ell_{2,1/4}$-norm. With 50% of labeled data, the best classification results on PIE and WarpPIE10P datasets are achieved by HSDAFS based on the $\ell_{2,1}$-norm. The KNN classifier achieves the highest accuracy with features selected by HSDAFS based on the $\ell_{2,1/4}$-norm on the UMIST and WarpAR10P datasets, and the $\ell_{2,1/2}$-norm on the COIL20 and ISOLET datasets.

Table 6 demonstrates the results of RF classifier obtained by different FS methods on all datasets.

We observe from Table 6 that when 10% of data are labeled, the RF classifier achieves the best classification results using the features selected by HSDAFS based on the $\ell_{2,1/2}$-norm on the COIL20, PIE, and UMIST datasets. Similarly, the best classification results of the RF classifier on the PIE dataset are attained by HSDAFS based on the $\ell_{2,1/2}$-norm. Furthermore, the proposed HSDAFS based on the $\ell_{2,1}$-norm exhibits the best performance on the WarpPIE10P and ISOLET datasets, and based on the $\ell_{2,1/4}$-norm on the WarpAR10P dataset. With 25% of labeled data, the RF classifier achieves the highest accuracy using the proposed HSDAFS based on the $\ell_{2,1}$-norm on the UMIST, WarpAR10P, and WarpPIE10P datasets. Additionally, the best classification results of the RF classifier on the COIL20, PIE, and ISOLET datasets are obtained using the proposed HSDAFS based on the $\ell_{2,1/2}$-norm, $\ell_{2,1/4}$-norm, and $\ell_{2,3/4}$-norm, respectively. When 50% of data are labeled, the proposed HSDAFS based on the $\ell_{2,1/4}$-norm and $\ell_{2,1/2}$-norm outperforms other FS methods using the RF classifier on the COIL20 dataset. Moreover, the best performance of the RF classifier on the PIE, WarpAR10P, and ISOLET datasets is achieved using the proposed HSDAFS based on the $\ell_{2,1/4}$-norm. Additionally, the RF classifier attains the highest accuracy using the features selected by HSDAFS based on the $\ell_{2,3/4}$-norm on the UMIST dataset and the $\ell_{2,1}$-norm on the WarpPIE10P dataset.

Table 5
Classification accuracy (Mean ± Std) of KNN classifier using different FS methods on different datasets.

L%	Method	COIL20	PIE	UMIST	WarpPIE10P	WarpAR10P	ISOLET
10%	FSLG	0.7269 ± 0.0184	0.3417 ± 0.0146	0.5992 ± 0.0186	0.4286 ± 0.0420	0.2820 ± 0.1559	0.6517 ± 0.0091
	SFSS	0.7347 ± 0.0218	0.4243 ± 0.0293	0.5973 ± 0.0393	0.4656 ± 0.0367	0.3334 ± 0.0678	0.5802 ± 0.0227
	SRS^{3}FS ($\ell_{2,1/2}$)	0.7091 ± 0.0398	0.3593 ± 0.0133	0.5857 ± 0.0349	0.4815 ± 0.0557	0.3590 ± 0.1026	0.5908 ± 0.0226
	SRS^{3}FS ($\ell_{2,1}$)	0.7423 ± 0.0147	0.4257 ± 0.0081	0.6108 ± 0.0353	0.4921 ± 0.0275	0.3504 ± 0.0534	0.5908 ± 0.0378
	Sr-SemiDFS	0.6906 ± 0.0431	0.2043 ± 0.0154	0.5684 ± 0.0372	0.4127 ± 0.0317	0.2479 ± 0.0783	0.4776 ± 0.0801
	SFS	0.7207 ± 0.0568	0.2130 ± 0.1449	0.5819 ± 0.0218	0.5026 ± 0.0875	0.2564 ± 0.0257	0.6698 ± 0.0347
	ASLCGLFS	0.7330 ± 0.0658	0.2830 ± 0.0531	0.5646 ± 0.0318	0.4497 ± 0.0783	0.2479 ± 0.0392	0.4584 ± 0.0317
	RDMRS2FS	0.6937 ± 0.0118	0.3250 ± 0.0278	0.5973 ± 0.0481	0.4815 ± 0.0331	0.2650 ± 0.0392	0.6325 ± 0.0061
	HSDAFS ($\ell_{2,1/4}$)	0.7554 ± 0.0185	0.4373 ± 0.0208	**0.6185 ± 0.0361**	0.4921 ± 0.0727	**0.3761 ± 0.0824**	0.6881 ± 0.0544
	HSDAFS ($\ell_{2,1/2}$)	0.7508 ± 0.0409	**0.4490 ± 0.0148**	0.6012 ± 0.0265	**0.5291 ± 0.0485**	0.2992 ± 0.0296	**0.7073 ± 0.0060**
	HSDAFS ($\ell_{2,3/4}$)	**0.7608 ± 0.0315**	0.4400 ± 0.0259	0.6146 ± 0.0241	**0.5291 ± 0.0601**	0.3419 ± 0.0740	0.6838 ± 0.0211
	HSDAFS ($\ell_{2,1}$)	0.7600 ± 0.0298	0.4400 ± 0.0168	0.6108 ± 0.0120	0.5079 ± 0.0420	0.2735 ± 0.0296	0.6763 ± 0.0529
25%	FSLG	0.8511 ± 0.0215	0.5667 ± 0.0070	0.8497 ± 0.0437	0.6561 ± 0.0601	0.3932 ± 0.0783	0.6752 ± 0.0393
	SFSS	0.8334 ± 0.0212	0.6083 ± 0.0378	0.8594 ± 0.0089	0.6825 ± 0.1299	0.5214 ± 0.1036	0.7511 ± 0.0619
	SRS^{3}FS ($\ell_{2,1/2}$)	0.8619 ± 0.0254	0.5860 ± 0.0632	0.8806 ± 0.0233	0.6614 ± 0.0330	0.5727 ± 0.0392	0.7308 ± 0.0363
	SRS^{3}FS ($\ell_{2,1}$)	0.8480 ± 0.0469	0.5987 ± 0.0623	0.8652 ± 0.0033	0.6931 ± 0.1322	0.5385 ± 0.0925	0.7564 ± 0.0544
	Sr-SemiDFS	0.7840 ± 0.0081	0.2533 ± 0.0127	0.8613 ± 0.0100	0.6297 ± 0.0903	0.3163 ± 0.0901	0.5022 ± 0.0514
	SFS	0.8665 ± 0.0058	0.6137 ± 0.0392	0.8343 ± 0.0241	0.7196 ± 0.0814	0.4530 ± 0.0391	0.7468 ± 0.0016
	ASLCGLFS	0.8457 ± 0.0094	0.5550 ± 0.0942	0.8362 ± 0.0167	0.6243 ± 0.0485	0.3675 ± 0.0824	0.5460 ± 0.0770
	RDMRS2FS	0.8588 ± 0.0080	0.5173 ± 0.0319	0.8747 ± 0.0318	0.6614 ± 0.0957	0.4274 ± 0.0296	0.7137 ± 0.0242
	HSDAFS ($\ell_{2,1/4}$)	0.8835 ± 0.0302	0.6587 ± 0.0021	**0.9037 ± 0.0146**	0.7725 ± 0.0243	0.5812 ± 0.0296	0.7511 ± 0.0045
	HSDAFS ($\ell_{2,1/2}$)	**0.8928 ± 0.0093**	**0.6600 ± 0.0070**	0.8748 ± 0.0371	0.7460 ± 0.0635	**0.6154 ± 0.0513**	0.7585 ± 0.0091
	HSDAFS ($\ell_{2,3/4}$)	0.8781 ± 0.0048	0.6560 ± 0.0070	0.8996 ± 0.0230	0.7355 ± 0.0642	0.5555 ± 0.0824	0.7650 ± 0.0151
	HSDAFS ($\ell_{2,1}$)	0.8920 ± 0.0035	0.6537 ± 0.0175	0.8940 ± 0.0133	**0.7884 ± 0.0330**	0.5043 ± 0.0148	**0.7885 ± 0.0182**
50%	FSLG	0.8989 ± 0.0127	0.6860 ± 0.0516	0.9306 ± 0.0321	0.9101 ± 0.0400	0.7265 ± 0.0148	0.7832 ± 0.0015
	SFSS	0.9322 ± 0.0116	0.7597 ± 0.1069	0.9364 ± 0.0152	0.9257 ± 0.0186	0.7265 ± 0.0148	0.8280 ± 0.0136
	SRS^{3}FS ($\ell_{2,1/2}$)	0.9090 ± 0.0766	0.7030 ± 0.0312	0.9538 ± 0.0058	0.9418 ± 0.0367	0.7265 ± 0.0148	0.8141 ± 0.0303
	SRS^{3}FS ($\ell_{2,1}$)	0.9367 ± 0.0142	0.7747 ± 0.0811	0.9557 ± 0.0120	0.9418 ± 0.0330	0.7350 ± 0.0296	0.8291 ± 0.0121
	Sr-SemiDFS	0.8156 ± 0.1093	0.2623 ± 0.0242	0.9268 ± 0.0088	0.8572 ± 0.0275	0.4530 ± 0.0391	0.5448 ± 0.0634
	SFS	0.9182 ± 0.0505	0.7953 ± 0.0391	0.9364 ± 0.0116	0.9100 ± 0.0330	0.6923 ± 0.0678	0.8301 ± 0.0197
	ASLCGLFS	0.9267 ± 0.0141	0.4717 ± 0.0630	0.9287 ± 0.0234	0.9047 ± 0.0275	0.5043 ± 0.0148	0.6015 ± 0.0016
	RDMRS2FS	0.9359 ± 0.0328	0.7480 ± 0.0763	0.9383 ± 0.0033	0.9418 ± 0.0330	0.6154 ± 0.0257	0.8216 ± 0.0197
	HSDAFS ($\ell_{2,1/4}$)	0.9583 ± 0.0023	0.8093 ± 0.0229	**0.9711 ± 0.0058**	0.9418 ± 0.0092	**0.7436 ± 0.1333**	0.8409 ± 0.0076
	HSDAFS ($\ell_{2,1/2}$)	**0.9622 ± 0.0081**	0.8080 ± 0.0248	0.9653 ± 0.0116	0.9630 ± 0.0242	0.6838 ± 0.0534	**0.8708 ± 0.0257**
	HSDAFS ($\ell_{2,3/4}$)	0.9599 ± 0.0174	0.8127 ± 0.0132	0.9615 ± 0.0120	0.9524 ± 0.0420	0.6496 ± 0.0296	0.8430 ± 0.0136
	HSDAFS ($\ell_{2,1}$)	0.9560 ± 0.0101	**0.8157 ± 0.0152**	0.9634 ± 0.0088	**0.9683 ± 0.0159**	0.6752 ± 0.0592	0.8419 ± 0.0181

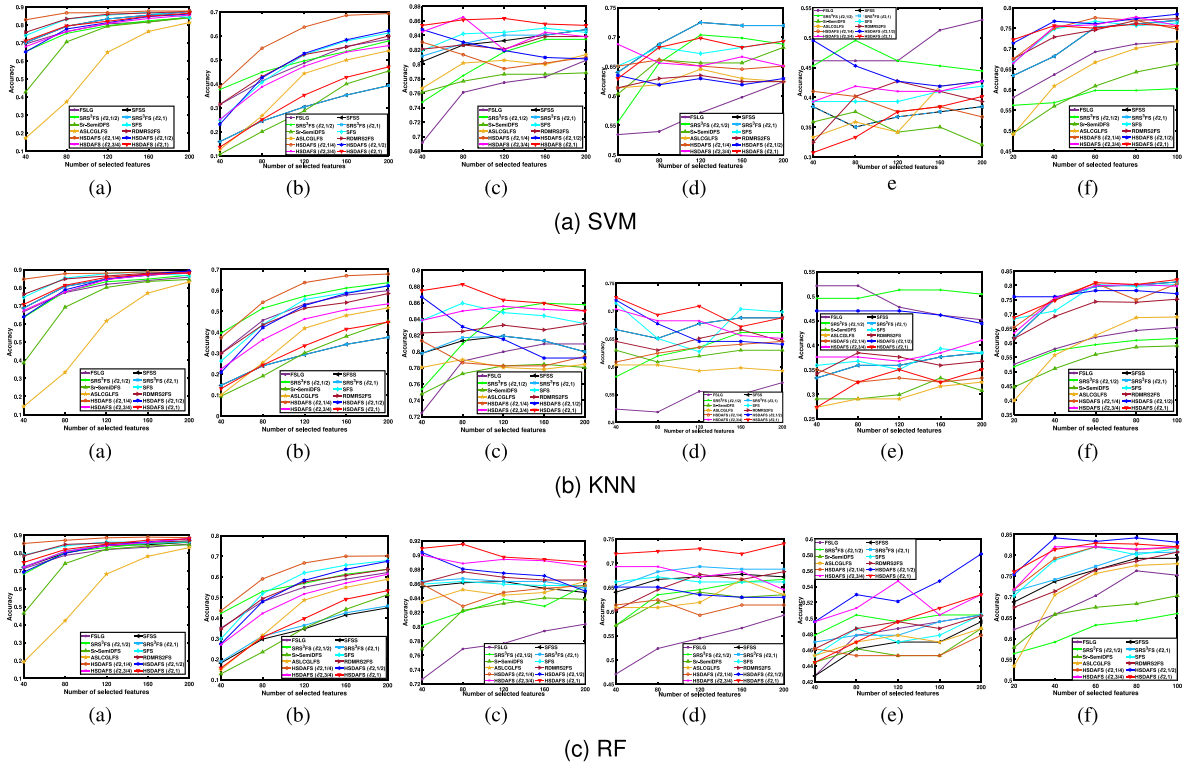


Fig. 1. The classification accuracy of different classifiers with various numbers of features on all datasets, including (a) COIL20, (b) PIE, (c) UMIST, (d) WarpPIE10P, (e) WarpAR10P, and (f) ISOLET.

Table 6
Classification accuracy (Mean ± Std) of RF classifier using different FS methods on different datasets.

L%	Method	COIL20	PIE	UMIST	WarpPIE10P	WarpAR10P	ISOLET
10%	FSLG	0.7122 ± 0.0158	0.4063 ± 0.0121	0.5626 ± 0.0121	0.4815 ± 0.0814	0.3333 ± 0.1175	0.6037 ± 0.0015
	SFSS	0.7114 ± 0.0186	0.4170 ± 0.0165	0.5819 ± 0.0590	0.4974 ± 0.0511	0.2992 ± 0.0148	0.5919 ± 0.0302
	SRS^{3}FS ($\ell_{2,1/2}$)	0.7230 ± 0.0149	0.4273 ± 0.0061	0.6070 ± 0.0352	0.5344 ± 0.4921	0.2992 ± 0.0645	0.5769 ± 0.0151
	SRS^{3}FS ($\ell_{2,1}$)	0.7253 ± 0.0140	0.4187 ± 0.0180	0.5954 ± 0.0417	0.5344 ± 0.0601	0.3248 ± 0.0296	0.5951 ± 0.0348
	Sr-SemiDFS	0.6783 ± 0.0418	0.2523 ± 0.0153	0.5279 ± 0.0261	0.4974 ± 0.0970	0.2308 ± 0.0679	0.4701 ± 0.0513
	SFS	0.7246 ± 0.0341	0.4217 ± 0.0245	0.5992 ± 0.0491	0.5132 ± 0.0661	0.3162 ± 0.0296	0.5941 ± 0.0121
	ASLCGLFS	0.7022 ± 0.0577	0.3210 ± 0.0507	0.5915 ± 0.0437	0.4709 ± 0.0733	0.2479 ± 0.1067	0.4797 ± 0.0075
	RDMRS2FS	0.7030 ± 0.0879	0.3633 ± 0.0268	0.6127 ± 0.0361	0.5026 ± 0.0733	0.2393 ± 0.0534	0.5374 ± 0.0468
	HSDAFS ($\ell_{2,1/4}$)	0.7407 ± 0.0324	0.4407 ± 0.0344	0.6281 ± 0.0426	0.5450 ± 0.0642	**0.3590 ± 0.0444**	0.6165 ± 0.0287
	HSDAFS ($\ell_{2,1/2}$)	**0.7423 ± 0.0241**	**0.4423 ± 0.0180**	**0.6416 ± 0.0417**	0.5224 ± 0.0180	0.3504 ± 0.0392	0.6442 ± 0.0045
	HSDAFS ($\ell_{2,3/4}$)	0.7238 ± 0.0232	0.4377 ± 0.0179	0.6397 ± 0.0203	0.5450 ± 0.0400	0.3504 ± 0.0296	0.6357 ± 0.0197
	HSDAFS ($\ell_{2,1}$)	0.7323 ± 0.0355	0.4413 ± 0.0134	0.6320 ± 0.0297	**0.5608 ± 0.0783**	0.3419 ± 0.0592	**0.6693 ± 0.0339**
25%	FSLG	0.8441 ± 0.0058	0.6200 ± 0.0425	0.8574 ± 0.0234	0.6667 ± 0.0159	0.5556 ± 0.0296	0.7500 ± 0.0212
	SFSS	0.8619 ± 0.0177	0.6547 ± 0.0216	0.8709 ± 0.0088	0.7037 ± 0.0330	0.5641 ± 0.0444	0.7607 ± 0.0030
	SRS^{3}FS ($\ell_{2,1/2}$)	0.8681 ± 0.0047	0.6097 ± 0.0248	0.8632 ± 0.0438	0.7037 ± 0.0184	0.5983 ± 0.0784	0.7479 ± 0.0121
	SRS3^{F}S ($\ell_{2,1}$)	0.8642 ± 0.0036	0.6617 ± 0.0187	0.8709 ± 0.0120	0.7143 ± 0.0420	0.6068 ± 0.0824	0.7746 ± 0.0045
	Sr-SemiDFS	0.8025 ± 0.0114	0.3037 ± 0.0103	0.8266 ± 0.0231	0.6455 ± 0.0485	0.4359 ± 0.0256	0.6186 ± 0.0619
	SFS	0.8534 ± 0.0075	0.6113 ± 0.0140	0.8745 ± 0.0319	0.7090 ± 0.0184	0.5385 ± 0.0678	0.7671 ± 0.0030
	ASLCGLFS	0.8457 ± 0.0142	0.5877 ± 0.0505	0.8632 ± 0.0186	0.6561 ± 0.0242	0.4530 ± 0.0148	0.6785 ± 0.0831
	RDMRS2FS	0.8488 ± 0.0109	0.4667 ± 0.0817	0.8767 ± 0.0120	0.7090 ± 0.0092	0.5043 ± 0.0392	0.7585 ± 0.0060
	HSDAFS ($\ell_{2,1/4}$)	0.8850 ± 0.0088	**0.6860 ± 0.0139**	0.8979 ± 0.0261	0.7249 ± 0.0092	0.6154 ± 0.0769	0.8119 ± 0.0030
	HSDAFS ($\ell_{2,1/2}$)	**0.8858 ± 0.0116**	0.6800 ± 0.0216	0.9094 ± 0.0261	0.6931 ± 0.0557	0.6239 ± 0.0645	0.8141 ± 0.0061
	HSDAFS ($\ell_{2,3/4}$)	0.8611 ± 0.0452	0.6640 ± 0.0338	0.8979 ± 0.0348	0.7143 ± 0.0572	0.6068 ± 0.1156	**0.8248 ± 0.0121**
	HSDAFS ($\ell_{2,1}$)	0.8704 ± 0.0242	0.6643 ± 0.0362	**0.9152 ± 0.0067**	**0.7355 ± 0.0242**	**0.6410 ± 0.0769**	0.8131 ± 0.0257
50%	FSLG	0.9236 ± 0.0040	0.7860 ± 0.0160	0.9248 ± 0.0153	0.9153 ± 0.0400	0.6838 ± 0.0148	0.8515 ± 0.0166
	SFSS	0.9221 ± 0.0128	0.8140 ± 0.0157	0.9268 ± 0.0120	0.9154 ± 0.0641	0.7521 ± 0.0296	0.8675 ± 0.0061
	SRS^{3}FS ($\ell_{2,1/2}$)	0.9352 ± 0.0080	0.8040 ± 0.0050	0.9364 ± 0.0252	0.9207 ± 0.0692	0.7350 ± 0.0784	0.8568 ± 0.0091
	SRS^{3}FS ($\ell_{2,1}$)	0.9025 ± 0.0590	0.8210 ± 0.0066	0.9364 ± 0.0116	0.9365 ± 0.0420	0.7521 ± 0.0645	0.8707 ± 0.0045
	Sr-SemiDFS	0.8534 ± 0.0669	0.3390 ± 0.0486	0.9307 ± 0.0153	0.9101 ± 0.0601	0.6752 ± 0.0148	0.7116 ± 0.0302
	SFS	0.9252 ± 0.0127	0.8090 ± 0.0461	0.9326 ± 0.0066	0.9259 ± 0.0330	0.7265 ± 0.0148	0.8430 ± 0.0196
	ASLCGLFS	0.9259 ± 0.0223	0.5440 ± 0.0606	0.9384 ± 0.0219	0.9153 ± 0.0558	0.6667 ± 0.0257	0.7062 ± 0.0197
	RDMRS2FS	0.9344 ± 0.0142	0.7710 ± 0.0795	0.9306 ± 0.0058	0.9471 ± 0.0641	0.7008 ± 0.0148	0.8291 ± 0.0091
	HSDAFS ($\ell_{2,1/4}$)	**0.9521 ± 0.0109**	**0.8470 ± 0.0113**	0.9557 ± 0.0120	0.9312 ± 0.0661	**0.8120 ± 0.0296**	**0.9030 ± 0.0103**
	HSDAFS ($\ell_{2,1/2}$)	**0.9521 ± 0.0117**	0.8413 ± 0.0160	0.9615 ± 0.0167	0.9418 ± 0.0485	0.7180 ± 0.0678	0.8718 ± 0.0091
	HSDAFS ($\ell_{2,3/4}$)	0.9506 ± 0.0158	0.8263 ± 0.0152	**0.9634 ± 0.0186**	0.9682 ± 0.0275	0.7607 ± 0.0645	0.8868 ± 0.0091
	HSDAFS ($\ell_{2,1}$)	0.9483 ± 0.0132	0.8283 ± 0.0290	0.9576 ± 0.0120	**0.9735 ± 0.0183**	0.7692 ± 0.0444	0.8932 ± 0.0030

4.4. Experimental influence of selected features

We evaluate the performance of various FS methods with different numbers of selected features. We use 25% of labeled training data and set the regularization parameters to 1. Fig. 1 illustrates the classification results of each FS method as the number of selected features varies. From Fig. 1, it is clear that the classification accuracy generally improves as the number of features increases. Additionally, it is evident that the proposed HSDAFS outperforms other FS methods across various numbers of features and classifiers in most cases. The highest classification accuracies on the COIL20 and WarpAR10P datasets are achieved with 200 features selected by HSDAFS based on $\ell_{2,1/2}$-norm using KNN and RF classifiers, respectively. On the PIE and WarpPIE10P dataset, the RF classifier attains the best accuracy using 200 features selected by HSDAFS based on the $\ell_{2,1/4}$-norm and $\ell_{2,1}$-norm, respectively. Similarly, on the UMIST dataset, the RF classifier achieves the highest classification result with 80 features selected by HSDAFS based on $\ell_{2,1}$-norm. Lastly, on the ISOLET dataset, the RF classifier exhibits the best performance using 40 and 80 features selected by the proposed HSDAFS based on $\ell_{2,1/2}$-norm.

4.5. Ablation study

To validate the effectiveness of the hypergraph Laplacian and $\ell_{2,p}$-norm ($0 < p \leq 1$) regularization in the proposed trace ratio-based FS method, we conduct ablation experiments. These experiments evaluate the performance of our trace ratio-based method using both the KNN hypergraph Laplacian and the KNN graph Laplacian with a KNN classifier on 25% of the labeled data. Additionally, we eliminate the regularization term to examine the performance of the KNN graph Laplacian-based SDA for FS and to assess the impact of regularization on the performance of the model. Table 7 compares the performance of the proposed FS method using the KNN hypergraph Laplacian (HSDAFS) and the KNN graph Laplacian (SDAFS), as well as the performance without the regularization term (SDA).

As shown in Table 7, the best results on all datasets are achieved by the proposed hypergraph Laplacian-based FS method. The experimental results demonstrate the advantage of using the hypergraph Laplacian in improving model performance. This improvement is due to the ability of the hypergraph to capture higher-order relationships among multiple data points simultaneously, a capability that a simple KNN graph cannot achieve. While a KNN graph considers pairwise relationships, a hypergraph can connect more than two data points within a single hyperedge, allowing it to represent complex interactions and group structures more effectively. The results also confirm the effectiveness of $\ell_{2,p}$-norm ($0 < p \leq 1$) regularization in enhancing the performance of the FS method. The ablation experiments reveal that both the hypergraph Laplacian and $\ell_{2,p}$-norm ($0 < p \leq 1$) regularization play crucial roles in enhancing the performance of the proposed method.

4.6. Convergence study

In this section, we conduct several experiments to empirically demonstrate the convergence of Algorithm 1. For these experiments, we use 10% of labeled data for each dataset, and the regularization parameters are fixed at 10. Fig. 2 illustrates the convergence behavior of the objective value in Eq. (20) using Algorithm 1 with different values of parameter p.

We can see from Fig. 2 that the value of Eq. (20) converges within a few iterations using Algorithm 1. This demonstrates the efficiency of our proposed algorithm. Moreover, from Fig. 2, we observe an association between the convergence rate of the proposed algorithm

Table 7
Comparison of the proposed FS method using the hypergraph Laplacian (HSDAFS), the graph Laplacian (SDAFS), and without the regularization term (SDA).

Method	COIL20	PIE	UMIST	WarpPIE10P	WarpAR10P	ISOLET
SDA	0.3542 ± 0.0152	0.1380 ± 0.0206	0.7303 ± 0.0561	0.6085 ± 0.1148	0.3248 ± 0.1648	0.3366 ± 0.0105
SDAFS ($\ell_{2,1/4}$)	0.8727 ± 0.0295	0.6267 ± 0.0110	0.8709 ± 0.0291	0.6984 ± 0.0159	0.5043 ± 0.0674	0.7372 ± 0.0211
SDAFS ($\ell_{2,1/2}$)	0.8766 ± 0.0071	0.6220 ± 0.0056	0.8574 ± 0.0089	0.6931 ± 0.0331	0.5385 ± 0.0870	0.7372 ± 0.0484
SDAFS ($\ell_{2,3/4}$)	0.8642 ± 0.0135	0.6360 ± 0.0173	0.8729 ± 0.0100	0.7037 ± 0.0092	0.4359 ± 0.0513	0.7479 ± 0.0000
SDAFS ($\ell_{2,1}$)	0.8811 ± 0.0013	0.6387 ± 0.0167	0.8825 ± 0.0089	0.6972 ± 0.0318	0.4801 ± 0.0148	0.7575 ± 0.0317
HSDAFS ($\ell_{2,1/4}$)	0.8835 ± 0.0302	0.6587 ± 0.0021	**0.9037 ± 0.0146**	0.7725 ± 0.0243	0.5812 ± 0.0296	0.7511 ± 0.0045
HSDAFS ($\ell_{2,1/2}$)	**0.8928 ± 0.0093**	**0.6600 ± 0.0070**	0.8748 ± 0.0371	0.7460 ± 0.0635	**0.6154 ± 0.0513**	0.7585 ± 0.0091
HSDAFS ($\ell_{2,3/4}$)	0.8781 ± 0.0048	0.6560 ± 0.0070	0.8996 ± 0.0230	0.7355 ± 0.0642	0.5555 ± 0.0824	0.7650 ± 0.0151
HSDAFS ($\ell_{2,1}$)	0.8920 ± 0.0035	0.6537 ± 0.0175	0.8940 ± 0.0133	**0.7884 ± 0.0330**	0.5043 ± 0.0148	**0.7885 ± 0.0182**

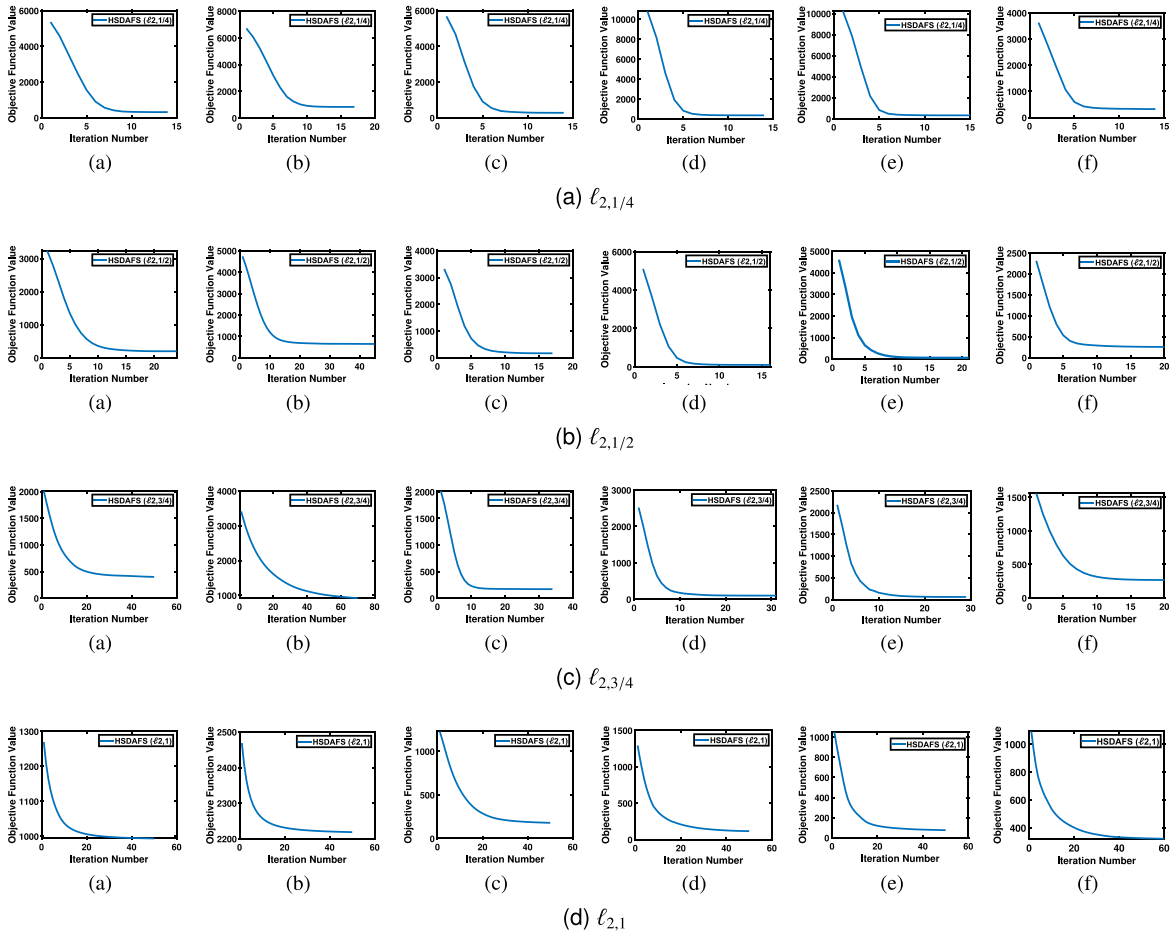


Fig. 2. Convergence behavior of our algorithm based on different values of parameter p on all datasets, including (a) COIL20, (b) PIE, (c) UMIST, (d) WarpPIE10P, (e) WarpAR10P, and (f) ISOLET.

and the value of parameter p. In most cases, the closer the value of parameter p is to zero, the faster the convergence, while the closer the value of parameter p is to one, the slower the convergence.

4.7. Non-parametric statistical analysis

To compare the proposed HSDAFS method with other FS methods and evaluate the significance of their classification differences, we employed a non-parametric Friedman test. This test ranks FS methods based on classification performance, assuming all methods perform equally under the null hypothesis. Rejection of this hypothesis indicates significant differences. With degrees of freedom ($df = k - 1$), where k is the number of FS methods, and a significance level of $\alpha = 0.05$, the critical chi-square value is 19.68 for $df = 11$ [37]. The calculated Friedman statistics for SVM, KNN and RF classifiers, based to the results of Tables 4–6 are 119.786325, 141.861111 and 150.58547, respectively. All calculated values exceed the critical chi-square value, leading to the rejection of the null hypothesis at the 0.05 level. Fig. 3 indicates the ranks of FS methods, with lower ranks indicating better performance. The HSDAFS method, using various norms, consistently achieves better ranks compared to other FS methods.

For post-hoc analysis, Holm's procedure was applied to evaluate statistical significance, with the best-ranked HSDAFS variant across each classifier serving as the control method. Holm's procedure rejects the null hypothesis when the p-value is less than Holm's adjusted value. Table 8 illustrates pairwise comparisons, indicating that the null hypothesis is rejected for all comparisons involving SVM and RF classifiers, and except for the comparison with the SRS^{3}FS ($\ell_{2,1}$) method in the case of the KNN classifier. These results highlight significant differences between HSDAFS and other FS methods.

We also conducted the Friedman statistical test for the ablation study results presented in Table 7. With a significance level of $\alpha = 0.05$ for 9 methods ($df = 8$), the critical chi-square value is 14.07. The

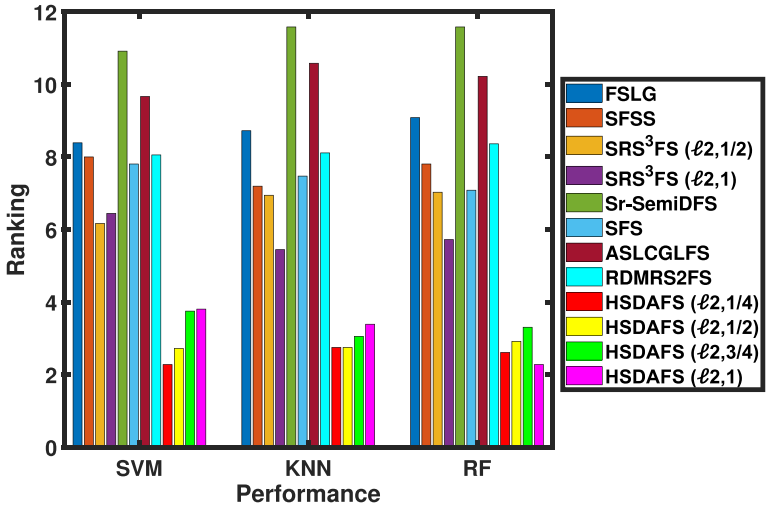


Fig. 3. Average ranks obtained by the Friedman test for each FS method across various classifiers.

Table 8
Post hoc comparisons on all classifiers with a significance level of $\alpha = 0.05$.

Method (SVM)	p (SVM)	Holm (SVM)	Method (KNN)	p (KNN)	Holm (KNN)	Method (RF)	p (RF)	Holm (RF)
Sr-SemiDFS	0	0.004545	Sr-SemiDFS	0	0.004545	Sr-SemiDFS	0	0.004545
ASLCGLFS	0	0.005	ASLCGLFS	0	0.005	ASLCGLFS	0	0.005
FSLG	0	0.005556	FSLG	0.000001	0.005556	FSLG	0	0.005556
RDMRS2FS	0.000002	0.00625	RDMRS2FS	0.000008	0.00625	RDMRS2FS	0	0.00625
SFSS	0.000002	0.007143	SFS	0.000085	0.007143	SFSS	0.000004	0.007143
SFS	0.000004	0.008333	SFSS	0.000217	0.008333	SFS	0.000064	0.008333
SRS^{3}FS ($\ell_{2,1}$)	0.000527	0.01	SRS^{3}FS ($\ell_{2,1/2}$)	0.000483	0.01	SRS^{3}FS ($\ell_{2,1/2}$)	0.000077	0.01
SRS^{3}FS ($\ell_{2,1/2}$)	0.001213	0.0125	SRS^{3}FS ($\ell_{2,1}$)	0.024967	0.0125	SRS^{3}FS ($\ell_{2,1}$)	0.004158	0.0125
HSDAFSS ($\ell_{2,1}$)	0.203662	0.016667	HSDAFSS ($\ell_{2,1}$)	0.595012	0.016667	HSDAFS ($\ell_{2,3/4}$)	0.392461	0.016667
HSDAFS ($\ell_{2,3/4}$)	0.220589	0.025	HSDAFS ($\ell_{2,3/4}$)	0.799312	0.025	HSDAFS ($\ell_{2,1/2}$)	0.595012	0.025
HSDAFS ($\ell_{2,1/2}$)	0.711531	0.05	HSDAFS($\ell_{2,1/2}$)	1	0.05	HSDAFS ($\ell_{2,1/4}$)	0.781511	0.05

Table 9
Average rankings obtained by the Friedman test for the proposed FS method using the hypergraph Laplacian (HSDAFS), the graph Laplacian (SDAFS), and without the regularization term (SDA).

Method	Ranking
SAD	9
SDAFS ($\ell_{2,1/4}$)	6.6667
SDAFS ($\ell_{2,1/2}$)	6.9167
SDAFS ($\ell_{2,3/4}$)	6.5
SDAFS ($\ell_{2,1}$)	5.1667
HSDAFS ($\ell_{2,1/4}$)	2.5
HSDAFS ($\ell_{2,1/2}$)	2.3333
HSDAFS ($\ell_{2,3/4}$)	3.1667
HSDAFS ($\ell_{2,1}$)	2.75

Table 10
Post hoc comparison for the proposed FS method using the hypergraph Laplacian (HSDAFS), the graph Laplacian (SDAFS), and without the regularization term (SDA) with a significance level of $\alpha = 0.05$.

Method	p	Holm
SAD	0.000025	0.00625
SDAFS ($\ell_{2,1/2}$)	0.003746	0.007143
SDAFS ($\ell_{2,1/4}$)	0.006132	0.008333
SDAFS ($\ell_{2,3/4}$)	0.008408	0.01
SDAFS ($\ell_{2,1}$)	0.07314	0.0125
HSDAFS ($\ell_{2,3/4}$)	0.598161	0.016667
HSDAFS ($\ell_{2,1}$)	0.792147	0.025
HSDAFS ($\ell_{2,1/4}$)	0.916051	0.05

calculated Friedman statistic is 37.211111, which exceeds the critical chi-square value, leading to the rejection of the null hypothesis at the 0.05 level. Table 9 presents the ranks of the proposed FS method using the hypergraph Laplacian (HSDAFS), the graph Laplacian (SDAFS), and without the regularization term (SDA). The HSDAFS method, using various norms, consistently achieves better ranks compared to the SDAFS method with various norms and the SDA method.

Subsequently, Holm's procedure was employed for post-hoc analysis to investigate statistical significance, with the HSDAFS ($\ell_{2,1/2}$) method, which has the best rank, serving as the control method. The results of the post-hoc comparison are illustrated in Table 10. The pairwise comparisons show that the null hypothesis is rejected for all comparisons involving SDAFS and SDA, except for the comparison with the SDAFS ($\ell_{2,1}$) method, indicating significant differences between HSDAFS and both SDAFS and SDA methods.

4.8. Running time

In Section 3.3, we have analyzed the computational complexity of the proposed HSDAFS method. Moreover, the computational complexity of various FS methods is outlined in Table 2. In this section, we perform experiments on COIL20, PIE, UMIST, and WarpPIE10P datasets to evaluate the running time of our proposed method and other FS methods. We utilize 10% of labeled data for each dataset in these experiments. It is worth noting that the experiments were executed using MATLAB R2018b on a computing system equipped with Windows 11, featuring a 1.50-GHz i7-1065G7 CPU and 16 GB of RAM. Table 11 presents the average time required by different FS methods.

As indicated in Table 11, the proposed HSDAFS based on $\ell_{2,1/4}$-norm requires more running time than FSLG, SFSS, and SRS^{3}FS based

Table 11
Running time (in seconds) of different FS methods.

Method	COIL20	PIE	UMIST	WarpPIE10P
FSLG	0.86	2.59	0.49	2.18
SFSS	0.95	3.02	4.36	27.62
SRS3FS ($\ell_{2,1/2}$)	0.79	2.45	0.43	2.39
SRS3FS ($\ell_{2,1}$)	0.92	2.98	3.44	24.99
SFS	18.44	38.24	16.77	188.49
ASLCGLFS	256.87	809.80	53.27	21.26
RDMRS2FS	18.76	38.31	15.33	81.12
HSDAFS ($\ell_{2,1/4}$)	10.22	32.19	10.14	132.14
HSDAFS ($\ell_{2,1/2}$)	25.64	43.17	11.96	176.49
HSDAFS ($\ell_{2,3/4}$)	28.65	52.99	17.79	290.21
HSDAFS ($\ell_{2,1}$)	36.95	55.71	22.34	297.20

on $\ell_{2,1/2}$-norm and $\ell_{2,1}$-norm, while requiring less running time than SFS, ASLCGLFS, RDMRS2FS, and HSDAFS based on $\ell_{2,1/2}$-norm, $\ell_{2,3/4}$-norm, and $\ell_{2,1}$-norm. Moreover, it is observed from Table 11 that the running time of the proposed method for the non-convex cases is frequently shorter compared to the convex case.

4.9. Overall results of the experiments

We can observe from the experiments that KNN and RF classifiers achieve the best performance on the COIL20 and WarpPIE10P datasets, respectively, when utilizing the top 100 features selected by the HSDAFS method based on $\ell_{2,1/2}$-norm with 50% of labeled data. Additionally, the highest accuracy on the UMIST dataset is achieved by the KNN classifier utilizing 100 features selected by the HSDAFS based on $\ell_{2,1/4}$-norm with 50% of labeled data. Moreover, using the RF classifier with 50% labeled data, the proposed HSDAFS based on $\ell_{2,1/4}$-norm achieves the best classification results with 100 features on the PIE and WarpAR10P datasets, and with 50 features on the ISOLET dataset. Furthermore, the results from Tables 4–6 indicate that the accuracies of various classifiers, when employing features selected by various FS methods, typically improve as the number of labeled data increases.

The results obtained from the experiments, as depicted in Fig. 1, reveal the significant impact of the selected features. It is evident that the proposed HSDAFS, with varying numbers of features, outperforms other FS methods across the majority of datasets. Furthermore, the results from Tables 4–6 demonstrate that the highest accuracies of all classifiers on all datasets are attained by the proposed HSDAFS method, emphasizing its efficacy in selecting the most discriminative features. This superiority can be attributed to several factors: Firstly, the HSDAFS method employs LDA in a semi-supervised manner. Secondly, it considers the correlation between features using the mixed $\ell_{2,p}$-norm to jointly select the most discriminative features. Lastly, it utilizes hypergraph Laplacian to preserve the high-order relationships within the data. By leveraging the geometrical structure of both labeled and unlabeled data using hypergraph Laplacian, along with the discriminative information present in the labeled data, the HSDAFS method demonstrates effective performance.

The results of ablation experiments demonstrate that the hypergraph Laplacian provides significant advantages in capturing high-order relationships within the data compared to the KNN graph Laplacian. Additionally, the $\ell_{2,p}$-norm ($0 < p \leq 1$) regularization enhances FS and improves learning performance.

We can also observe from the results that different values of parameter p in the HSDAFS lead to different results. Moreover, based on the experimental results, it is evident that in some cases, the performance of HSDAFS based on $\ell_{2,1}$-norm surpasses that of HSDAFS based on $\ell_{2,p}$-norm ($0 < p < 1$), and vice versa. Consequently, determining the optimal value of p in the proposed $\ell_{2,p}$-norm ($0 < p \leq 1$) FS method cannot be definitively established and relies on the characteristics of the data.

5. Conclusion and future research

In this paper, we proposed a sparse FS method called HSDAFS, formulated in the trace ratio form, utilizing hypergraph Laplacian-based SDA. In the proposed HSDAFS method, the $\ell_{2,p}$-norm ($0 < p \leq 1$) is applied to the trace ratio form of the objective function of the hypergraph Laplacian-based SDA, making it suitable for FS. The HSDAFS method leverages the geometrical structure of both labeled and unlabeled data based on hypergraph Laplacian and regularized discriminative FS to jointly select discriminative features from all data. Here are the main advantages of our method: (1) Preservation of high-order relationships of data: The utilization of hypergraph Laplacian in our method preserves the geometrical structure of all data, allowing for the capture of high-order relationships among data points. (2) Utilization of discriminative capabilities in a semi-supervised manner for sparse FS: Leveraging the discriminative information of labeled data in the proposed $\ell_{2,p}$-norm-based SDA method for FS enables the identification of the most discriminative features. The $\ell_{2,p}$-norm regularization within the hypergraph Laplacian-based SDA method ensures that the correlations between features are considered during the selection of discriminative features.

The mixed $\ell_{2,p}$-norm transforms our trace ratio-based method into a convex problem when $p = 1$ and a non-convex one when $0 < p < 1$. Consequently, optimizing our method becomes complex. Hence, we converted the proposed trace ratio-based method into an equivalent trace difference method and proposed an efficient algorithm capable of uniformly solving the HSDAFS method for all $p \in (0, 1]$. Theoretical and experimental proofs have demonstrated that our algorithm converges to the optimum solution. One limitation of the HSDAFS method is that the optimal value of p in the proposed $\ell_{2,p}$-norm FS method cannot be definitively determined and is dependent on the characteristics of the data. Extensive experiments were performed on several datasets to evaluate the HSDAFS method. The results indicated the effectiveness and superiority of the HSDAFS method compared with other semi-supervised sparse FS methods.

Although the proposed HSDAFS method performs well, it can still be further improved in several areas. Currently, in our method, equal weights are assigned to each hyperedge during the construction of the hypergraph. However, different hyperedges may contain different information about the data structure. Therefore, a future direction will focus on learning different weights for different hyperedges. Additionally, our current approach constructs the hypergraph Laplacian matrix separately from the FS process. Due to the presence of noise and outliers in the original data, the fixed hypergraph Laplacian matrix constructed from the data may be unreliable. Hence, another future work will involve developing a sparse feature selection method based on semi-supervised discriminant analysis, which adaptively constructs a hypergraph Laplacian matrix within the FS framework.

CRediT authorship contribution statement

Razieh Sheikhpour: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Kamal Berahmand:** Writing – original draft, Validation, Resources, Methodology, Investigation. **Mehrnoush Mohammadi:** Writing – original draft, Validation, Methodology, Conceptualization. **Hassan Khosravi:** Writing – review & editing, Validation, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this paper are public datasets, and the GitHub link has been included within the paper.

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